Farthest vertices and dimension bounds for compact hyperbolic Coxeter polytopes

Coxeter Polytopes Frontier Project

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Abstract

We obtain a dimension bound of 23 for compact hyperbolic Coxeter polytopes from estimates at a vertex farthest from an interior point. One common spatial vector has squared norm less than one, and its orthogonal projections onto the finite vertex components have additive costs. Actual opposite walls, positive acute projections, and feasible timelike combinations give stronger constraints than edge lengths alone. Every A_r, $r \geq 7$, costs more than $1/2$, and every A_r, $r \geq 17$, costs more than $5/7$. A localized prefix estimate supplies the complementary cost for large B and D components. A stronger A_{20} estimate and a coupled D_{23}–A_1 argument close the last two patterns. The component estimates allow arbitrary other finite components at the chosen vertex, so their costs can be combined in one budget. Arithmeticity is not required. The argument uses actual walls and feasible points of the original polytope throughout.

1 Statement and proof strategy

A compact hyperbolic Coxeter polytope has finitely many facets and dihedral angles π/m, $m \geq 2$. Its outward unit facet normals lie in Lorentz space of signature $(n, 1)$, and the polytope is given by $(q, n_i) \leq 0$ on the future unit hyperboloid.

Theorem 1.1. *Every compact hyperbolic Coxeter polytope has dimension at most 23.*

The published starting bound is $n \leq 29$ [1]. We exclude dimensions 24 through 29 by examining one actual finite vertex. The external geometric and diagram inputs are compact acute simplicity, the nonpositive products of distinct outward wall normals, finite spherical Coxeter classification, and the fact that a Lannér diagram has at most five vertices; see [2]. All component estimates and component exclusions used below are internal arguments.

Sections 2–8 give the common geometry, ordinary costs and debits, complete finite continuation lists, actual component pressures, prefix argument, and global deduction. The component propositions are proved here rather than attributed to the published classification. Appendix A gives a guide to the supporting derivations and the scope of the accompanying verification material.

The method does not require an arbitrary facet to be a Coxeter polytope. It also does not replace actual walls by a convenient abstract Gram completion. Every auxiliary point below satisfies the inequalities of the entire original polytope.

1.1 The common argument and its case splits

The proof starts with one geometric observation, rather than a list of diagrams. At a farthest vertex, every finite component constrains the orthogonal projection of the same vector x. These squared

projection norms add, and their sum is less than one. The task is therefore to prove component costs that no sufficiently large finite forest can afford.

The argument first separates at the component type. Small and exceptional types admit the ordinary costs of Proposition 3.1. Large A components require a coupled analysis of the two ends of a chain, carried out in Section 5. Large B and D components instead admit the coordinate-prefix estimates of Section 6. These estimates allow arbitrary other components at the chosen vertex; a bound for a whole irreducible vertex would not suffice.

The branches reunite in Section 8. Its rank and energy accounting excludes dimensions 25 through 29 and leaves only two patterns in dimension 24. Those two patterns explain the last case-specific arguments: the stronger A_{20} cost in Section 5.5, and the coupling between D_{23} and its short isolated spectator in Section 7. They are parts of this proof, rather than separate dimension-bound papers.

The finite continuation lists used here are mathematical case analyses. Each needs a proof that its allowed contacts are exhaustive and that its rejected branches cannot occur. A source hash checks neither assertion. The proof uses no numerical search or combinatorial enumeration script: the finite lists are justified by the retained finite diagrams and explicit affine witnesses before their energy inequalities are used.

2 One actual vector at a farthest vertex

Fix a strict interior future unit point z of a compact polytope P. The function $q \mapsto -(z,q) = \cosh d(z,q)$ is strictly convex on every geodesic, so a maximum over P occurs at a vertex p. Write

$$z = E(p - x), \qquad E = -(z,p) > 0, \qquad x \perp p.$$

Then

$$\|x\|^2 = 1 - E^{-2} < 1. \tag{1}$$

The incident normals have a positive definite Gram matrix B. Let $C = B^{-1}$, let $\omega_i = \sum_j C_{ij} n_j$ be the spatial duals, and put

$$d_i = C_{ii}, \qquad u_i = \omega_i / \sqrt{d_i}.$$

The finite diagram is an orthogonal union of irreducible components. Thus $x = \sum_T x_T$ and $\|x\|^2 = \sum_T \|x_T\|^2$.

The inward direction of edge i is $-u_i$. If its length is L_i, its other endpoint is $p_i = \cosh L_i \, p - \sinh L_i \, u_i$. Farthest comparison gives

$$x \cdot u_i \geq \tau_i := \tanh(L_i/2) > 0. \tag{2}$$

Strict interiority also gives $x \cdot n_i > 0$. For an actual outside unit wall write

$$c = Ap - v, \qquad A = -(p,c) > 0, \qquad \beta_i = -(c, n_i) \geq 0.$$

Its spatial vector is $v = \sum_i \beta_i \omega_i$, with $A^2 = \beta^T C \beta - 1$, and

$$x \cdot v < A. \tag{3}$$

2.1 Positive projections and reciprocal shadows

A positive definite matrix with nonpositive off-diagonal entries has entrywise nonnegative inverse. If J is a positive principal set of actual normals, projecting a strict interior timelike vector onto $J^{\perp}$ adds a nonnegative combination of those normals. The result remains timelike and strictly satisfies every other wall inequality. Hence a positive principal set defines an actual face. In particular, a positive full replacement containing an already actual wall gives an actual finite endpoint.

An affine principal set is impossible in a compact chamber. Its positive kernel gives a nonzero future null combination with nonpositive products against every wall. Nonzeroness follows by pairing with a strict interior point. This would be a feasible ideal point. These arguments use the all-wall sign in [2, p. 42, §2.1].

Lemma 2.1 (Actual reciprocal shadow)**.** *For an actual outside wall c, put $v_i = (C\beta)_i/\sqrt{d_i}$ and $T_i = v_i/A$. If $T_i > 1$, the wall c is the actual opposite of edge i and $\tanh L_i = 1/T_i$. If it is not that opposite, then $\tanh L_i \geq T_i$. The value $T_i = 1$ is impossible in the compact case. Distinct actual opposites satisfy*

$$\tanh L_i \, \tanh L_j \geq K_{ij} := \frac{C_{ij}}{\sqrt{d_i d_j}}. \tag{4}$$

Proof. The Schur identity says that $C - C_{\cdot i}C_{i\cdot}/d_i$ is zero on row and column i and equals the inverse of the retained old Gram elsewhere. It is entrywise nonnegative. If a different actual opposite has height D and contacts γ, acuteness therefore gives

$$AD \geq \beta^T C\gamma \geq \frac{(C\beta)_i (C\gamma)_i}{d_i}.$$

Its endpoint equation is $\tanh L_i = D\sqrt{d_i}/(C\gamma)_i$, proving the alternative. For $T_i > 1$ that alternative is impossible, while the positive replacement criterion gives the stated actual closure. Equality gives an ideal rather than finite endpoint. Applying the same Schur inequality to the actual opposite of i gives $T_j \geq K_{ij}/\tanh L_i$, and proves (4). □

It is essential that (4) uses full tangents. Write $F(t) = 2t/(1+t^2)$. If a component has energy at most $e < 1$, then for a pair of correlation k,

$$F(\tau_i)F(\tau_j) \leq \frac{4s}{(1+s)^2}, \qquad s = \frac{e(1+k)}{2}. \tag{5}$$

Indeed $\tau_i + \tau_j \leq \sqrt{2e(1+k)}$, and $\log F$ is concave on $(0,1)$. A strict reverse comparison with k forces the same actual opposite.

Lemma 2.2 (Opposites respect whole components)**.** *Two orthogonal outgoing directions cannot share an actual opposite at a farthest vertex.*

Proof. Suppose i, j share c and $u_i \cdot u_j = 0$. Clear the other old normals positively, obtaining c_F in the span of p, u_i, u_j. Strict interiority gives $(z, c_F) < 0$. Since both endpoints lie on c, with $r_i = \tanh L_i$ and $r_j = \tanh L_j$,

$$c_F = Ap - \frac{A}{r_i}u_i - \frac{A}{r_j}u_j, \qquad \frac{x \cdot u_i}{r_i} + \frac{x \cdot u_j}{r_j} < 1.$$

But (2) makes the left side at least $1 + (\tau_i^2 + \tau_j^2)/2 > 1$, a contradiction. □

In each retained foreign component an actual opposite has at most one contact: two would give a cycle in a finite diagram. Its cost is $q = \beta_j^2 C_{jj} \geq 1/4$. In a nontrivial component, $C_{jj} \geq 4/3$, so there is no cost in $(1/4, 1/3)$. Moreover

$$q < A^2. \tag{6}$$

Otherwise Lemma 2.1 closes the contacted foreign direction too, contrary to Lemma 2.2, or gives a null endpoint in the equality case.

2.2 Actual finite and timelike core points

For an old set J let $w = \sum_{j \in J} (B_J^{-1} \beta_J)_j n_j$ and $\delta = 1 - \beta_J^T B_J^{-1} \beta_J$. Then w has nonnegative coefficients and $\eta = c + w$ has square δ, products zero on J, and product δ with c.

If $\delta < 0$, η is future timelike and satisfies every actual wall inequality. Farthest comparison with $\eta/\sqrt{-\delta}$ gives

$$x \cdot (v - w) \geq A - \sqrt{-\delta}. \tag{7}$$

For $\delta > 0$, the positive acute projection onto $J + c$ gives

$$x \cdot (v - w) \geq H(A, \delta) := \frac{A}{1 + \sqrt{\delta/(A^2 + \delta)}} > A - \sqrt{\delta}. \tag{8}$$

For example the projected point before normalization is $p + (A/\delta)(c + w)$. Its products with $J + c$ vanish; all other inequalities follow from positivity. These constructions may retain all old foreign components, and never infer a new wall from a formal Gram matrix.

More generally any feasible future point $\xi = Ap - U$ of square $-k < 0$ gives

$$x \cdot U \geq A - \sqrt{k}, \qquad \|x_{\operatorname{supp} U}\|^2 \geq \frac{A - \sqrt{k}}{A + \sqrt{k}}. \tag{9}$$

Here $\|U\|^2 = A^2 - k$. The support notation refers to actual orthogonal components, not to a projected mirror assumed native.

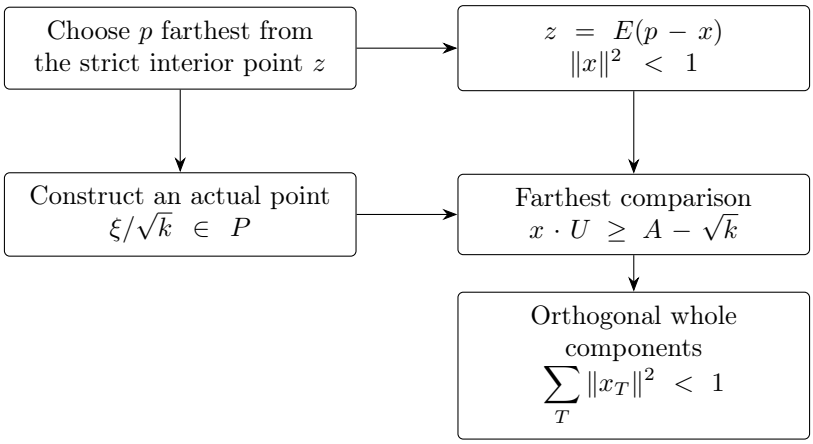


Figure 1: The common-vector argument. Every component cost is tested on the same x. Feasibility of the auxiliary point is proved against all actual walls before its pressure is used.

3 Finite-component costs and the two small debits

We call a block ordinary when its actual energy is strictly greater than its rank divided by 14. The following internal estimates provide the complete finite-type coverage needed later.

Proposition 3.1 (Ordinary component interface)**.** *The bounds in Table 1 hold with arbitrary finite spectators at a farthest vertex. There is at most one short A_1 direction, where short means $L < \operatorname{arsinh}(1/\sqrt{2})$. Every other A_1 pays more than $1/10$. Apart from that possible direction, the only nonordinary small blocks are the two isolated rank-two defects described below, each still paying more than $9/100$.*

Type	Strict energy lower bound	Type	Strict energy lower bound
A_2	319/2000, with one exception	$I_2(m)$, $m \geq 4$	1/7, with one exception
A_3, B_3, H_3	1/4	A_4	2/7
B_4, D_4, F_4	1/3	H_4	11/36
A_5	3/8	A_6	3/7
B_5, B_6	3/7	B_7, B_8	1/2, 4/7
D_5, D_6	5/14, 3/7	D_7, D_8	1/2, 4/7
E_6, E_7, E_8	9/20, 5/9, 2/3		

Table 1: Internal ordinary rates. Every displayed nonexceptional rate is at least rank/14, with a strict energy inequality.

We prove the proposition in this section. The proofs use the common geometry above and finite spherical classification; foreign components are retained in every positive replacement. In the contact lists, $m@j$ means contact $\cos(\pi/m)$ at old node j, and a missing contact means zero. Put $h = A^2$, and write the own and total foreign costs of a wall as q and Q, so that $h = q + Q - 1 > 0$.

Lemma 3.2 (Common-wall forcing and isolation)**.** *Suppose the energy of a component is at most e. A pair for which the right side of (5) is below K_{ij} has one actual opposite. If one wall closes every direction of the component, then, with $v_i = (C\beta)_i/\sqrt{d_i}$ and $R_e^2 = 4e/(1+e)^2$,*

$$q - 1 \leq h \leq R_e^2 \min_i v_i^2. \tag{10}$$

If $h < 1/4$, it has no foreign contact. In a path, a wall closing every direction can have multiple contacts only at the two endpoints; in a D_4 star it has at most one.

Proof. The first assertion is (5). For a closed direction, $\tanh L_i = A/v_i$ and $\tau_i^2 \leq e$, which gives (10). Every nonzero foreign cost is at least $1/4$ and is strictly below h by (6); hence the isolation assertion. If the old path between two contacts omits a direction closed by the wall, deleting that direction leaves a cycle in an actual finite replacement. This is impossible. The same argument applies to every two contacts in a D_4 star. Finally, a single-contact own cost $q = 1$ gives an affine principal set, impossible even in the presence of further spectators. □

3.1 The short directions and their exact debits

For a short A_1 opposite put $b = \beta^2$, $h = A^2$, and let Q be its full foreign cost. Then

$$h = b + Q - 1, \qquad h < b/3, \qquad Q < 1 - 2b/3.$$

Equation (6) rules out two foreign components. With one, its cost must be the sole value below $1/3$, namely an A_1 contact 3 of cost $1/4$. Thus the only possibilities are

$$I:\ \text{isolated dotted } A_1,\quad 0<h<1/2; \qquad J:\ \text{dotted } A_1+A_{1,3},\quad 1/3<h<3/8.$$

For completeness, with one foreign component, $Q<h=b+Q-1$ forces $b>1$, and $Q<1/3$ then forces $Q=1/4$. For J, the heavy edge contributes more than $A/2$ to $x\cdot v$. The light shadow contributes at least $A(1-\sqrt{1-1/(4h)})$; this is at least $A/2$ if $h\le 1/3$. Strict (3) therefore gives the asserted lower bound. Native acuteness excludes two short walls. Indeed their height product is below $1/2$, so their product must be zero. Disjoint supports give a negative product, heavy overlaps give a spatial product above $1/2$, and a sole light/light overlap of two J walls gives $1/4$, below their height product. None is zero. A J heavy direction has $\tanh^2 L>4/13$, hence energy $>9/100$; its light direction has $\tanh^2 L>2/3>16/25$, hence energy $>1/4$. Thus J funds both A_1 factors above their combined ordinary baseline $2/10$. Every nonshort direction pays at least $5-2\sqrt{6}>1/10$, since $\sqrt{6}<49/20$. We may conservatively discard the one short direction, but only the isolated type I can create an unfunded debit.

The rank-two exceptions are

old block / new isolated contact	$h=A^2$	energy
$A_2/7$	$(4/3)\cos^2(\pi/7)-1\in(0,1/12)$	$>9/100$
$I_2(7)/3$	$(1/4)\csc^2(\pi/7)-1\in(0,1/3)$	$>9/100.$

(11)

Their spatial supports lie entirely in their own blocks. Two defects in different blocks, or a defect and a short A_1, would have actual wall product $-\sqrt{h_1h_2}\in(-1/2,0)$. That interval is forbidden: distinct acute unit walls have product 0 or at most $-1/2$. Therefore

$$\text{at most one short } A_1,\quad \text{or one rank-two defect, but never both.} \tag{12}$$

This argument is independent of every other component type.

3.2 The two rank-two estimates

For A_2, set $e_2=319/2000$ and $\sinh^2 L_0=31/50$. If both edges have length at least L_0, then their half-tangents exceed $173/500$ and energy is greater than $4\cdot173^2/(3\cdot500^2)>e_2$. A shorter edge cannot have two positive own contacts: $3/3$ is affine, while $3/4$ or larger gives $\sinh^2 L\ge 4\sqrt{2}/9>31/50$, increasing with either contact and with Q. If only the other node is contacted, apply the same test to that shorter contacted direction. For one contact of square b, the shorter-edge condition is

$$h=4b/3+Q-1<124b/243, \qquad Q<1-200b/243. \tag{13}$$

Every positive foreign cost must be below h. A label 3 gives $h<1/4$, hence isolation and negative height. For a label 4, $h<62/243$ forces each foreign cost to be $1/4$; the only positive possibility is $Q=1/2$, but then $h=1/6<1/4$. A label 5 has $Q<1/2$, hence at most one foreign component, whose cost exceeds h because its own cost $\varphi^2/3$ is below one. A label 6 is affine. For a finite label $m\ge 7$, $b>.81$ gives $Q<1/3$; thus $Q=0$ or $Q=1/4$. The latter is possible in (13) only for $b<729/800$, namely $m=7,8,9,10$. Indeed $\cos^2(\pi/11)>45/49>729/800$, using $\pi<22/7$. A dotted contact has $Q<1/4$ and is isolated.

For an isolated finite contact the same wall closes the mate, since its reciprocal ratio squared is $b/(4b-3)>1$. For $m\ge 8$ that actual mate has $\tanh^2 L\ge 6\sqrt{2}-8>400/841$, so its energy exceeds $4/25>e_2$. For an isolated dotted contact, $b<243/200$ and its distinct mate has shadow

$\tanh^2 L \geq b/(4b-3) > 81/124 > 16/25$. For a mixed $m = 7,8,9,10$ contact, the mate's white cost is $b + 1/4 > 1$; the distinct shadow instead gives $\tanh^2 L \geq b/(4b - 9/4) > 11/17 > 16/25$. Both give energy greater than $1/4$. Thus the only exception is isolated new 7 at old A_2. The root $t = \cos(2\pi/7)$ of $8t^3 + 4t^2 - 4t - 1$ lies in $(623/1000, 5/8)$: the polynomial changes sign there and is increasing on $[1/2, 1]$. Hence $b > 1623/2000$ and its actual mate satisfies $\tanh^2 L = 4 - 3/b > 164/541 > 3600/11881$, proving its cost $> 9/100$.

For an old $I_2(m)$, $m \geq 4$, put $k = \cos(\pi/m)$ and $d = (1-k^2)^{-1}$, so $C = d\left(\begin{smallmatrix} 1 & k \\ k & 1 \end{smallmatrix}\right)$. At energy at most $1/7$, (5) forces one common opposite. Two contacts give an actual $\cosh L \geq \cot(\pi/(2m))/\sqrt{3} > 11/8$, with individual energy $> 3/19 > 1/7$. For one contact β,

$$q(1 - 7k^2/16) \leq 1, \qquad q = d\beta^2. \tag{14}$$

Its minimum left side is $(9d+7)/64$. Since $m \geq 8$ gives $d \geq 4 + 2\sqrt{2} > 19/3$, only four old labels remain:

m	consequence of (14)
4	$\beta^2 \leq 16/25 < \cos^2(\pi/5)$: 3, 4; $h \leq 7/32 < 1/4$, $q \leq 1$;
5	$d > 25/9$: only 3; $h < 7/32$, $q < 1$;
6	only 3; $q = 1$;
7	$d > 5$: only 3; $Q < 3/16$.

Isolation or the affine principal excludes the first three rows. The last is isolated new 3 at old $I_2(7)$, with $0 < h < 1/3$. Its actual mate has $\cosh L = \cot(\pi/7)/\sqrt{3} > 109/91$ by the same cubic bracket; the squared comparison has positive numerator 2652. Thus its energy also exceeds $9/100$, completing (11) and (12).

3.3 Rank three and four: one common-wall calculation

For energy $1/4$, adjacent correlation at least $1/\sqrt{3}$ forces a common opposite: the lowest cleared comparison is $74/\sqrt{3} > 39$. At energy $1/3$ the threshold $1/\sqrt{2}$ works, since $51/\sqrt{2} > 34$. For A_4 at energy $2/7$, the adjacent correlations are $\sqrt{3/8}, 2/3, \sqrt{3/8}$; the forcing test is $k^3 + 16k^2 + 36k - 28 > 0$, whose lowest value is positive because $291\sqrt{6} > 704$. Direct inversion verifies the indicated correlation thresholds for A_3, B_3, H_3 and B_4, D_4, F_4. For H_4 use energy $11/36 < 1/3$. Thus a common wall closes all directions in every case.

Number path bonds as $A_3 : (3,3)$, $B_3 : (4,3)$, $H_3 : (5,3)$, $A_4 : (3,3,3)$, $B_4 : (4,3,3)$, $F_4 : (3,4,3)$, and $H_4 : (5,3,3)$. By Lemma 3.2, multiple contacts can only be at the endpoints. Two endpoint 3 contacts in A_3, A_4 are affine; the next minimum is $3/4$. In the other rows below even $3/3$ is excluded. The displayed values are lower bounds for $q - R_e^2 v_j^2$, contradicting (10):

T	j	lower bound
A_3	2	$177/200 + 9\sqrt{2}/100 > 1$
A_4	2	$(230 + 15\sqrt{2})/243 > 1$
B_3	2	$(77 + 18\sqrt{2})/100 > 1$
B_4	3	$15/16 + \sqrt{2}/8 > 1$
F_4	2	$15/16 + \sqrt{2}/4 > 1$
H_3	2	$(72\varphi + 95)/100 > 1$.

D_4 already has at most one contact. In H_4, two endpoint contacts retain, after deleting old node 4, a path at least $(3,5,3)$ of determinant $(5 - 4\varphi)/16 < 0$. Hence all contacts are single.

For a single contact at j let $\sigma_j = \min_i(v_i^2/q)$. Formula (10) gives $q \leq (1 - R_e^2\sigma_j)^{-1}$. The complete inverse calculation is as follows; entries are in path order, and D_4 means leaf,center.

T	d_j	σ_j	upper q
A_3	$3/2, 2, 3/2$	$1/9, 1/3, 1/9$	$225/209, 75/59, 225/209$
A_4	$8/5, 12/5, 12/5, 8/5$	$1/16, 1/6, 1/6, 1/16$	$162/155, 243/215, 243/215, 162/155$
B_3	$3, 4, 2$	$1/3, 1/2, 1/3$	$75/59, 25/17, 75/59$
D_4	$2, 4$	$1/4, 1/2$	$16/13, 8/5$
B_4	$4, 6, 4, 2$	$1/4, 1/3, 1/2, 1/4$	$16/13, 4/3, 8/5, 16/13$
F_4	$4, 12, 12, 4$	$1/2, 2/3, 2/3, 1/2$	$8/5, 2, 2, 8/5$

Here R_e^2 is respectively 16/25, 56/81 for A_4, and 3/4 for the 1/3 rows. Dividing the upper q by d_j gives the finite native lists: A_3 endpoints 3/4/5, middle 3/4; A_4 endpoints 3/4, inner 3; B_3 nodes $3; 3; 3/4$; D_4 leaf 3/4, center 3; B_4 nodes $3; \varnothing; 3; 3/4$; F_4 endpoints 3, inner $\varnothing$. Each has $q \le 1$. Equality is affine; in every other listed case (10) gives $h < 1/4$ (the largest needed bounds are $16/75, 7/155, 28/405, 3/16$). Isolation then gives $h = q - 1 < 0$. This proves the six crystallographic and A rows.

For H_3, the minimum contact $\beta = 1/2$ at node 1 or 2 gives tests at node 3 equal to $(199\varphi + 263)/500$ and $(61\varphi + 77)/125$, both above one. At node 3, $\beta \ge 1/\sqrt{2}$ gives the test at node 1 at least $(27\varphi + 118)/150 > 1$. Only 3@3 remains, with $q = (\varphi + 2)/4 < 1$ and $h \le 4\varphi^4/75 < \varphi/4$. No spectator gives negative height; one has cost $Q > h$; two have $Q \ge 1/2$ and hence $h \ge q - 1/2 = \varphi/4$. All are impossible. For H_4, $R_e^2 = 1584/2209$; at $\beta = 1/2$ the four tests (contact node, test direction) are

$$(1,4) : \frac{4938\varphi + 3688}{2209}, \quad (2,4) : \frac{7209\varphi + 5334}{2209}, \quad (3,1) : \frac{2500\varphi + 2667}{2209}, \quad (4,1) : \frac{625\varphi + 1219}{2209}.$$

All exceed one, the last by $\varphi > 198/125$; larger contacts only increase these positive tests. This proves $H_4 > 11/36$.

3.4 The two intermediate A components

Use $C_{ij} = 2\min(i,j)(N - \max(i,j))/N$, with $N = 6, 7$. For A_5, distinct 2/3 and 3/4 pairs each demand a sum above 109/100. The vector $u_2 + 2u_3 + u_4$ has square $7 + 4\sqrt{2} < 443/35$, and

$$(109/50)^2 - (3/8)(443/35) = 209/35000 > 0.$$

For A_6, the three pair demands are

$$2249/2000, \qquad 11547/10000, \qquad 2249/2000;$$

the vector $u_2 + 2u_3 + 2u_4 + u_5$ has square $< 33781/1250$, with demand 34037/10000 and Cauchy gap 2181583/700000000 at energy 3/7. These pair bounds follow by substitution in $F(s/2)^2 < K$. Thus a common 2/3 occurs for A_5, and a common 2/3 or 3/4 for A_6, up to reversal.

Each retained path allows one contact. The full two-contact list is

Component/pair	positions	necessary finite-white labels
$A_5, 2/3$	$1, 3$	3/3 or 4@1 + 3@3
	$1, 4; 1, 5; 2, 4; 2, 5$	both 3
	$2, 3$	both in $3, 4, 5$
$A_6, 2/3$	$1, 3$	3/3 or 4@1 + 3@3
	$1, 4; 1, 5; 1, 6; 2, 4; 2, 5; 2, 6$	both 3
	$2, 3$	$3, 4, 5$ at 2; $3, 4$ at 3
$A_6, 3/4$	$3, 4$	both in $3, 4, 5$
	other eight left/right pairs	both 3

Indeed deletion of either closed node leaves a connected finite path or D/E_6 tree: a 4 can only be terminal, and D/E bonds are all 3. All-3 doubles are affine cycles. The adjacent non-3/3 doubles force an actual edge of cosh at least $(3+\sqrt{2})/2 > 11/5$ in A_5; for A_6 the minimum formulas are $(5+4\sqrt{2})/\sqrt{7}$, $(4\sqrt{2}+6)/\sqrt{14}$, and $(12+9\sqrt{2})/\sqrt{42}$, all above 5/2. The A_5 terminal-4 row has $Q=0$ and cosh $(3+\sqrt{2})/2$. The A_6 version has white cost 9/10, hence $Q=0$, and is treated below. These cosh thresholds imply individual energies above 3/8 and 3/7.

For single 3 contacts the complete own costs and strict white bounds are

component/pair	j	q	Q upper
$A_5, 2/3$	$1,2,3,4,5$	$5/12, 2/3, 3/4, 2/3, 5/12$	$2/3, 2/3, 5/8, 1/2, 5/8$
$A_6, 2/3$	$1,2,3,4,5,6$	$3/7, 5/7, 6/7, 6/7, 5/7, 3/7$	$2/3, 2/3, 3/5, 2/5, 2/5, 3/5$
$A_6, 3/4$	$1/6; 2/5; 3/4$	$3/7; 5/7; 6/7$	$5/8; 1/2; 5/8.$

End and small-Q rows have $h < 1/4$, or only one foreign component with cost $Q > h$. In A_5 at $j = 2,3$ two foreign components would each exceed $1/3, 1/4$, respectively, forcing $Q \geq 2/3$; the same gap treats $A_6, j = 2$. In the remaining $A_6, j = 3,4$ rows, two foreign bonds must be 3 since $Q < 5/8$; together with the contacted old node and its two neighbors they form affine two-fork D_5. This excludes all single 3 cases. A single internal 4 gives the affine $(3,3,4)$ star; an end 4 has $Q \leq 1/3$ and $h < 1/4$, again impossible.

A single 5 is at a retained path end; its H_3/H_4 white cost exceeds 3/4, so $Q = 0$. All cases are covered by these actual positive replacements:

component/pair	contact	actual direction	cosh L
$A_5, 2/3$	1 or 5	4 or 2	$\varphi^3/\sqrt{3}$
	2	3	φ^2
	3	2	$\sqrt{3}\varphi^3$
$A_6, 2/3$	1	4	$3\varphi^3/\sqrt{21}$
	2	3	$4\varphi^2/\sqrt{14}$
$A_6, 3/4$	1 or 6	4 or 3	$3\varphi^3/\sqrt{21}$
	3 or 4	4 or 3	$9\varphi^3/\sqrt{21}.$

The first group exceeds 11/5, the second 5/2, using $\varphi > 8/5$, $\sqrt{21} < 23/5$, $\sqrt{14} < 15/4$. No new wall is inferred: each replacement contains the already actual c. Finally the A_6 double $4@1 + 3@3$ has the positive actual future point

$$\xi = c + (\sqrt{2} - 1/2)n_1 + (\sqrt{2} - 1)n_2 + (\sqrt{2}/2)n_3, \qquad \xi^2 = 5/4 - \sqrt{2} = -k.$$

Its products with c, n_1, n_3 vanish and its product with n_2 is $(\sqrt{2}-3)/4 < 0$. Here $A^2 = (5+4\sqrt{2})/7$ and $A^2 - 9k = (335 - 236\sqrt{2})/28 > 0$; (9) gives energy $> (A - \sqrt{k})/(A + \sqrt{k}) > 1/2$.

3.5 The four intermediate B components

Use $u_i = (e_1 + \cdots + e_i)/\sqrt{i}$ and, for $i < r$, $\omega_i = \sqrt{2}(e_1 + \cdots + e_i)$, while $\omega_r = e_1 + \cdots + e_r$. At energies $3/7, 1/2, 4/7$ the relevant forcing signs are $127\sqrt{2/3} - 100$, $(69\sqrt{6/7} - 52)/7$, and $(199\sqrt{6/7} - 176)/7$, all positive. Thus B_5, B_6 have a common 2/3, and B_7, B_8 a common 6/7.

For B_5, B_6, deletion 2 retains $A_1 + B_3/B_4$. The right contact is 3@3, with the sole B_5 exception 3@5 giving F_4; deletion 3 allows one left 3/4/5. For 3@3, own 0 is affine, own 3 is a triangle, and own ≥ 4 gives cosh $L_3 \geq 2 + 3/\sqrt{2} > 3$. With no right contact the complete choices are 3/4/5 at 1 or 2: 3@2 and either 4 are affine; 3@1 has $h = Q - 1/2 < 1/6$ and closes a forbidden foreign

direction. The two 5 cases have $Q = 0$: 5@1 also closes 4 with $\cosh L_4 = \varphi^3 > 3$, and 5@2 has $\cosh^2 L_3 = 2\varphi^2/(2-\varphi) > 9$. For the B_5 short-end 3@5, finite F_4 forbids left 1 and all spectators; own 0 is affine and the other replacement forces own 3, giving $\cosh L_2 = 2 + \sqrt{2}$. Every branch exceeds the individual threshold 11/5.

For the 6/7 pair, a wall without a right contact has a single left 3 or endpoint 4. Apart from far 3@1 these give an affine fork, double-4, or star. A minimal nonpositive principal for the far branch must contain $c, 7$, since deletion 7 is finite; their connecting path $c, 1, \ldots, 7$ has eight vertices. Spectators attach through c and do not shorten it. This violates the Lannér order bound, excluding the branch with all spectators. In B_7, let β be the contact to the retained right A_1:

β/own 6	left contacts	pressure or rejection
$\beta \geq \cos(\pi/5)$, any	none	$\cosh L_6 \geq \sqrt{6}\cot(\pi/5) > 3$
$\beta_4, 0$	any allowed	affine $(4,4)$
$\beta_4, > 0$	none	$\cosh L_6 \geq \sqrt{24}$
$\beta_3, > 0$	none	$\cosh L_6 \geq 2 + \sqrt{2}$
$\beta_3, 0$	none or 1	affine F_4
$\beta_3, 0$	one $j = 2, \ldots, 5$	$\cosh L_6 \geq 1 + 3/\sqrt{2} > 3$.

Here $\beta_m = \cos(\pi/m)$. The last row retains the full foreign cost: its white residual is $\Delta \leq 3/4 - j(6-j)/12 \leq 1/3$, and $v_6 \geq \sqrt{12}(j/12 + 1/(2\sqrt{2}))$. Thus no allowed E_7 extension is silently removed.

In B_8 the retained right B_2 has one simple 3 at either end. At long end 7, own 0/3 is affine and own 4 gives $\cosh L_7 \geq 6 + 7/\sqrt{2} > 7$. At short end 8, own > 0 gives $\cosh L_6 \geq \sqrt{6} + \sqrt{3} > 11/3$. At own 0, the left-white budget permits only 3@1 or 3@5: none or 1 gives affine F_4, but 3@5 is an allowed bridge, with $\Delta \leq 1/12$ and $v_6 = \sqrt{12}(5/12 + 1/(2\sqrt{2}))$. It has $\cosh L_6 \geq 5 + 3\sqrt{2} > 9$; it is estimated, not removed by an incorrect F_4 sealing claim. These complete lists prove the four asserted B costs.

3.6 Extended stars, D_7, and the D_8 closure

Let S be a central D_4 star with an extended arm. If one wall closes its center and three neighbors, every minimal nonpositive principal contains c and all four nodes of S, since deletion of any of them is an actual finite replacement. Compactness and the Lannér order bound make that principal exactly $S + c$. Two contacts give a nonpositive proper triangle or cycle. A sole center 3 is affine, and a larger center contact has a nonpositive proper star. A sole leaf contact is 3/4/5 from its finite proper path: 3 gives finite D_5, 4 affine, leaving only 5. If that leaf extends, it retains a nonpositive $(5,3,3)$ star; if another arm extends, it retains a nonpositive $(5,3,3,3)$ path. Each omits a supposedly closed S direction, a contradiction. Further negative contacts cannot repair the positive nonpositive quadratic witness. Thus the four opposites cannot all coincide.

For D_5, D_6 the minimum central correlations are $\sqrt{3/5}, \sqrt{2/3}$. At energies $5/14, 3/7$ the forcing signs are $544\sqrt{3/5} - 362$ and $127\sqrt{2/3} - 100$, positive. For E_6, E_7, E_8 the arm lengths are $(1,2,L)$ with $L = 2,3,4$. In raw norm-two coordinates the central Schur residual is $\delta = 2 - [1/2 + 2/3 + L/(L+1)]$; on an arm of length m, squared central correlation is $t/(\delta + t)$, $t = m/(m+1)$. The minima are $\sqrt{3}/2, \sqrt{6/7}, \sqrt{15}/4$. At energies $9/20, 5/9, 2/3$ even $4e/(1+e)^2$ lies below these correlations. All four opposites would coincide, proving the five extended-star costs.

For D_7 write $o = r - 2$, $a = r - 3$, with fork leaves b, f. At energy at most 1/2, correlation $\sqrt{4/5}$ forces a common o, a. Deletion a retains $A_3 = b - o - f$, so its right contact is none, middle 3, or endpoint 3/4/5; deletion o allows one left contact. The complete surviving profiles are

$$4@a + 3@o, \qquad \text{isolated } 5@b, \qquad 3@a + 4@b, \tag{15}$$

up to exchanging the fork leaves. To check the list: middle 3 forces $4@a$ (own $0/3$ is affine); endpoint 5 seals all white extensions by H_4, and a contact at the removed a leaves a nonpositive $(3,3,3,5)$ path. Endpoint 4 forces $3@a$, otherwise an affine terminal-4 fork remains. Endpoint 3 excludes every left contact by cycles or retained F_4; a foreign 3 is affine E_7, a foreign 4 is an affine terminal-4 fork, and larger foreign bonds are nonfinite. Its isolated height is negative for $r = 7$ and it contains affine E_8 for $r \geq 8$. With no right contact, internal-left 3 and endpoint 4 are affine; far-left 3 needs a Lannér path of at least six vertices. This accounts for all spectators as well as the removed own contact. The first and third roles have $\cosh L_o = a + (r-2)/\sqrt{2} > 7$. The isolated 5 has $\tanh^2 L_a = 1 - (9 - 4\sqrt{5})/(r-3)$; at $r = 7$ its half-tangent square is $4/\sqrt{5} - 1 > 3/4$. Each role therefore contradicts energy at most $1/2$, proving the D_7 cost.

For D_8, common $4/5$ has exactly $4@4 + 3@5$. Indeed deletion 4 permits simple 3 at a right D_4 leaf, and deletion 5 allows one left 3 or endpoint 4. Long-leaf 3 forces $4@4$ and forbids foreign contacts by the retained $(3,4,3,3)$ path. A fork 3 with a left 3 gives an affine cycle; without a left contact it is affine E_8; a left 4 is nonpositive. With no right contact only far $3@1$ survives the affine tests, but its path to 5 has six vertices. Common $5/6$ has the preceding center-neighbor list. Both common pairs thus force an actual half-tangent square above $3/4$. At energy at most $4/7$, their opposites are distinct.

Here is the remaining three-direction calculation. Put $q = \sqrt{4/5}$, $k = \sqrt{5/6}$, $\lambda = (1+k)/2$, $\alpha = k\sqrt{1-k}/2$, $e = 4/7$, and $P(t) = t/(1+\sqrt{1-t})^2$. Write $h_i = x \cdot u_i$ and decompose $x = aw + bv + zt_0 + y$, where w, v are the normalized sum and difference of u_5, u_6, $u_4 = qu_5 + t_0/\sqrt{5}$, and $y \perp w, v, t_0$. The full-tangent laws imply

$$h_4 h_5 \geq P(q), \qquad T := (1+\alpha)b^2 + \lambda z^2 \leq \delta := \lambda e - P(k) < 1/400.$$

For the second inequality take the tangent line to $4p/k - (1+p)^2$ at its first zero $p = P(k)$, with $p = h_5 h_6 = \lambda a^2 - (1-\lambda)b^2$. Keeping the negative quadratic terms in $h_4 h_5$ gives

$$h_4 h_5 \leq q\lambda e + A_0|b| + B_0|z| - qkb^2 - q\lambda z^2, \qquad A_0^2 = 8/105, \quad B_0^2 = 4/35.$$

Weighted Cauchy bounds the linear part by $\sqrt{C_0 T}$, where $C_0 = (8/105)/(1+\alpha) + (4/35)/\lambda < 47/250$. Also $qk/(1+\alpha) > .71$, $q > .71$, and $q\lambda e < .489$. Consequently

$$h_4 h_5 < .489 + \sqrt{(47/250)T} - .71T < 101781/200000 < .509 < P(q),$$

a contradiction. The rational seals are $.8944 < q < .8945$, $.9128 < k < .9129$, $.29 < \sqrt{1-k} < .2953$, $.13 < \alpha < .14$, and $.95 < \lambda < .95645$. The tight δ comparison reduces to $91280000 \cdot 70000 - 167780209 \cdot 38083 = 26300653 > 0$; the last square-root comparison uses $\sqrt{47/100000} < 271/12500$. This proves $D_8 > 4/7$ and completes Proposition 3.1.

4 Two classical interfaces used in the composition

Proposition 4.1 (Actual B/D floor)**.** *Every whole B_r or D_r component with $r \geq 9$ has actual projection energy greater than $9/14$, with arbitrary finite spectators.*

Proof. Assume energy at most $e = 9/14$. In B it suffices to bound the first seven coordinates; all displacements below lie there. The normalized directions in either type are $u_i = (e_1 + \cdots + e_i)/\sqrt{i}$. We first establish distinctness of $4/5, 6/7$, then use those laws to exclude all common $1/2, 2/3$ profiles. This order is noncircular.

For an internal common $i/(i+1)$ with right retained rank at least four, finite classification permits a right long-leaf 3, or none. The first forces own 4: own $0/3$ is affine fork/triangle, and

any foreign contact retains the nonpositive $(3,4,3,3)$ path. Without a right contact, internal-left 3 and endpoint 4 are affine. Only far 3@1 remains, but its path to $i+1$ has at least six vertices, contradicting the Lannér bound. Right D-fork contacts give affine E_8 or a cycle with the left contact. Thus the sole common profile is $4@i + 3@(i+1)$, $Q = 0$, with $\cosh L_{i+1} = i + (i+1)/\sqrt{2} > 7$. The D_8 right-D_4 and center-neighbor boundary calculations are in Section 3; the center-neighbor calculation also handles D_9 at 6/7. In B_9, the extra right short-end 3@9 gives F_4, excluding left and foreign white contacts. The removed own 6 remains to check: own 0 is affine F_4, own 4 the affine $(4,3,3)$ star, and own 3 gives $\cosh L_6 = (1+1/\sqrt{2})\sqrt{12} > 11/2$ and $\tau_6^2 > 9/13 > e$. Consequently 4/5 and 6/7 are distinct.

Write $h_i = x \cdot u_i$. Distinct 6/7 gives $h_6 h_7 \geq k/(1+\sqrt{1-k})^2 > 9/16$, where $k = \sqrt{6/7} > 576/625$ with squared margin 21318. Hence $x \cdot w > 3/2$ for $w = u_6 + u_7$. We record four local tests with their exact margins.

Edge 2 *with* $\cosh L_2 \geq 1+\sqrt{2}$. Then $h_2^2 \geq \sqrt{2}-1$. Projection of w about u_2 has $\rho = 1/\sqrt{3} + \sqrt{2/7} < 1.114$ and residual square $S = 29/21 + 4\sqrt{2/21} < 2.62$. Its Cauchy envelope $\rho t + \sqrt{S(e-t^2)}$ decreases for $t \geq q = \sqrt{\sqrt{2}-1}$: its derivative condition follows from $\rho^2 e < 13/16 < 6/5 < \|w\|^2 q^2$. Since $q < .644$, $e - q^2 < .23$, and $\sqrt{2.62 \cdot .23} < .777$, $x \cdot w < 1.114(.644) + .777 = 1.494416 < 3/2$. These follow by squaring, using $1.414 < \sqrt{2} < 1.4143$, $1/\sqrt{3} < .578$, $\sqrt{2/7} < .535$, $\sqrt{2/21} < .309$.

Isolated 5@1. Its height square is $(\varphi - 1)/2$ and its actual direction-4 shadow has $\tau_4^2 = \varphi - 1$. For the same w, projection about u_4 has $\rho = 2(1/\sqrt{6} + 1/\sqrt{7}) < 787/500$ and $S = 16/21 + 4/\sqrt{42} < 69/50$. Use $1/\sqrt{6} < .409$, $1/\sqrt{7} < .378$, $\sqrt{42} > 162/25$; the last residual comparison has cleared margin 23. Since $\sqrt{e} < 401/500$, $e - (\varphi - 1) < 1/40$ and $(93/500)^2 > 69/2000$, Cauchy gives $x \cdot w < 362087/250000 < 3/2$.

Isolated 5@2. Its height square is φ. The direction-1 shadow gives $x_1 > 31/50$, since $F(31/50)^2 < 161/200 < \varphi/2$ with cleared margin 6541881. Its actual edge 3 has $\cosh L_3 = \sqrt{2}\varphi^2 > 37/10$; hence $S_3 = x_1 + x_2 + x_3 > 131/100$, using $3\tau_3^2 > 81/47 > (131/100)^2$, of margin 3433. Distinct 4/5 gives $x \cdot w_0 > 57/40$, $w_0 = u_4 + u_5$. Put $v = e_1 + e_2 + e_3$, $k_0 = 1/2 + 1/\sqrt{5} < 237/250$, and $x_0 = (31/50, 69/200, 69/200, 0, \ldots)$. Then $x_0 = (11/40)e_1 + (69/200)v$, $x \cdot x_0 > x_0^2 > 31/50$. The residual $w_0 - k_0 v$ has square $13/20 + 1/\sqrt{5} < 11/10$ and excess demand $> 9/50$. All coefficients of $x_0 + (w_0 - k_0 v)/6$ remain positive, and its Cauchy margin is at least

$$(31/50 + (9/50)/6)^2 - e(31/50 + (11/10)/36) = 3/700 > 0.$$

Contacts 4@2 + 3@3. The white A_2 cost is 2/3, so $Q = 0$ or 1/4; the latter has $\cosh L_3 = 4 + 3\sqrt{2} > 7$. At $Q = 0$, put $s = \sqrt{2}$. The actual floors satisfy $\tau_2^2 = (1+2s)/7$, $\tau_3^2 = (2s-1)/3$, $\tau_2\tau_3 = 1/\sqrt{3}$. Their positive equality vector has $E_0 = (6s-4)/7 > .6407$. The positive old-A_2 correction of c is an actual future point of square $-s/3$, giving pure displacement

$$U = \frac{2+s}{2}e_1 + \frac{1+s}{3}(e_2+e_3+e_4), \qquad x \cdot U \geq A - \sqrt{s/3}, \quad A^2 = (5+4s)/2.$$

Its projection coefficients on u_2, u_3 are $\rho_2 = (2s+1)/6$, $\rho_3 = (1+s)/\sqrt{3}$; its residual square is $(7+4s)/12 < 1.055$, with excess demand $\Delta > .06$. Indeed $1.4142 < s < 1.4143$ gives $A > 2.3083$, $\sqrt{s/3} < .6867$, $\tau_2 \in (.7395, .7396)$, $\tau_3 \in (.7806, .7808)$, $\rho_2 < .6381$, $\rho_3 < 1.394$, and excess $> .06122604$. The old positive coefficients exceed $.3, .52$; subtracting $(\rho_2, \rho_3)/16$ removes less than $.04, .088$. The corrected dual therefore has Cauchy margin at least

$$(.6407 + .06/16)^2 - e(.6407 + 1.055/256) = 8825113/11200000000 > 0.$$

This is an actual-point constraint, not an edge-only assertion.

Now exhaust common 1/2. Deletion 1 retains right rank at least eight, so the right contact is none or long-leaf 3@2. The latter forces own $\gamma \geq 1/\sqrt{2}$ (own 0/3 is affine), and $\cosh L_2 = (1+\gamma)/\sqrt{1-\gamma^2-Q} \geq 1+\sqrt{2}$, already excluded. Without a right contact, $A^2 = 2\gamma^2 + Q - 1 > 0$ and $\gamma^2 + Q < 1$. Own 0 has zero derivative; own 3 has height below 1/4 and forces a foreign closure; own 4/6 is affine. Own 5 with $Q > 0$ has $\cosh L_2 \geq \varphi^2 > 1+\sqrt{2}$; at $Q = 0$ use the isolated-5 test. Finite $m \geq 7$ forces $Q = 0$; $m \geq 8$ uses the edge-2 test, leaving only isolated 7.

For common 2/3, deletion 2 retains right B/D rank at least seven; deletion 3 allows one left 3/4/5 at 1 or 2. A right long 3@3 forces own 3/4/5: own 0 is affine, own 3 a triangle, own 4 is the tested $4+3$ profile, and own 5 has $Q = 0$, $\cosh L_3 = (2+5\varphi)/\sqrt{2} > 7$. Without a right contact the full left list is $3@1, 4@1, 5@1, 3@2, 4@2, 5@2$. The first has $A^2 = Q - 1/2 < 1/6$ and forces foreign closure; $4@1, 3@2, 4@2$ are respectively affine terminal-4, two-fork, and $(4,3,3)$ star. The 5 cases have $Q = 0$ by white cost $\varphi^2/3 > 3/4$ and were tested above. For D_9 one extra right-fork 3 gives retained $D_7 + c = E_8$. It excludes old-1 and foreign contacts, but the removed own 2 must still be checked: own 0/3/4 gives affine E_8/cycle/star, and own 5 retains nonpositive $(3,5,3)$. At higher rank the fork attachment already contains affine E_8. No finite branch is omitted.

For the final isolated 7@1, put $t = \cos(2\pi/7)$, $g = \cos(\pi/7)$, and

$$\xi = c + g^{-1}n_1 + (1-t)g^{-1}n_2 = \sqrt{t}\,p - \frac{V}{\sqrt{1+t}}, \qquad V = t(e_1+e_2) + (1-t)e_3.$$

It has $\xi \cdot c = \xi \cdot n_1 = 0$, $\xi \cdot n_2 = (1/2 - t)/g < 0$ and $\xi^2 = -(1-t)(2t-1)/(1+t) < 0$. All coefficients and its time coefficient are positive, proving actual feasibility and future orientation. Thus $x \cdot V \geq \sqrt{t(1+t)} - \sqrt{(1-t)(2t-1)}$. The increasing cubic $8t^3 + 4t^2 - 4t - 1$ gives $.6234 < t < .6235$, with endpoint values $-.000926392768, .000105223$. Hence the first radical is > 1.0059, the second $< .305$, and $.95 < \|V\| < .9588 < .959$, so $x \cdot \widehat{V} > .73$. The projection of w onto $\widehat{V}$ is $\rho = (1/\sqrt{6} + 1/\sqrt{7})(1+t)/\|V\| \in (1.3, 1.35)$, and residual square is $< 11/5$. The Cauchy envelope decreases above .73, since $\rho^2 e < 6561/5600 < 3(.73)^2 < \|w\|^2(.73)^2$. It gives $x \cdot w < 1.35(.73) + .492 = 591/400 < 3/2$; the final squared radical margin is $277/1750000$. Thus $1/2, 2/3, 6/7$ all have distinct actual opposites.

Their exact pair sums are

$$x \cdot (u_1 + u_2) > 1.091, \quad x \cdot (u_2 + u_3) > 1.265, \quad x \cdot (u_6 + u_7) > 1.512.$$

Set $W = .1(u_1 + u_2) + .055(u_2 + u_3) + .305(u_6 + u_7)$. Direct positive Gram estimates give

$$x \cdot W > .639835, \qquad W^2 < .636435825,$$
$$.639835^2 - \frac{9}{14}(.636435825) = \frac{70423123}{280000000000} > 0.$$

To verify the pair sums, use $F(a)F(b) \leq F((a+b)/2)^2$. The correlation lower bounds

$$7071/10000, \qquad 81649/100000, \qquad 9258/10000$$

have squared margins $959, 322397, 26052$ below the exact values. At sums $m/1000$, $m = 1091, 1265, 1512$, the cleared differences

$$p(4000000 + m^2)^2 - 16dm^2\,1000000$$

are respectively

$$40828209713231, \qquad 358399613480625, \qquad 52443946917888.$$

The six upper Gram entries of the three pair vectors are $3.415, 3.633, 3.852, 3.102, 1.899, 2.474$, obtained by positive squaring of their prefix correlations. Their expansion is the stated bound on W^2. This contradicts Cauchy and proves the proposition. □

Proposition 4.2 (Large-component caps)**.** *At a farthest vertex, with arbitrary finite spectators, the following components are impossible:*

$$A_r \ (21 \le r \le 29), \qquad B_r \ (23 \le r \le 29), \qquad D_r \ (24 \le r \le 29).$$

These are internal theorems. We give their finite continuation and energy proofs below. All references to old directions concern the chosen farthest vertex, and all added walls are already actual.

4.1 Clearing spectators and the finite continuation list

Put $c_0 = 61/100$. For a long index, write $D_i = \omega_i^2$, $q_i = \sqrt{D_i}$ and $h_i = x \cdot \omega_i$, and call it weak if $h_i \le q_i - c_0$. An actual edge with

$$\frac{\cosh L_i}{q_i} > T(q_i), \qquad T(q) = \frac{1 + (1 - c_0/q)^2}{q[1 - (1 - c_0/q)^2]} = \frac{1}{c_0} - \frac{q - c_0}{q(2q - c_0)},$$

is strong. We use $T(q) < 100/61$ and, on $5/2 \le q \le 31/8$, $T(q) < 761/500$. For the latter, clearing denominators gives the positive concave quadratic $-715800q^2 + 3268319q - 1860500$; check its two endpoints.

First consider an isolated own contact γ with full foreign cost Q. Write $v = \gamma\omega_i + v_S$, $u = (1 - Q)/\gamma^2 > 0$, so $A^2 = \gamma^2(D_i - u)$. If $u < 1$, the positive actual combination

$$\xi = c + v_S + \frac{1 - Q}{\gamma} n_i$$

has products zero with c, S, product $\gamma(u - 1) < 0$ with n_i, and square $-\gamma^2 u(1 - u)$. It is future and feasible, giving the pure component pressure

$$h_i > \sqrt{D_i - u} - \sqrt{u(1 - u)}. \tag{16}$$

At $u = 1$ the same nonzero feasible vector is null, impossible. For $u > 1$, the actual positive set $S + n_i + c$ has residual $\gamma^2(u - 1)$; (8) and $x \cdot n_i > 0$ give

$$h_i > \sqrt{D_i - u} - \sqrt{u - 1}. \tag{17}$$

In A the sharper form is $h_i > \sqrt{D_i - 1} - \sqrt{u - 1}$: with $B = \sqrt{D_i - 1}$, $t = \sqrt{u - 1}$ the exact finite pressure is $\sqrt{B^2 - t^2}/(1 + t/B) > B - t$. No normalized projected mirror is asserted native.

The own labels $3, 4, 6$ and parallel contact are excluded by the local affine fork/star, $(6, 3)$ path, and null sum. Thus $\gamma \ge \varphi/2$. If $Q > 0$, $u \le 3/\varphi^2 = 6 - 3\varphi < 7/6$. For A, $D_i \ge 85/11$, $q_i > 11/4$, and $\sqrt{D_i - 1} > 5/2$. Formula (17) has loss $< 4/21 + 5/12 = 17/28 < c_0$, since $u - 1 \le 5 - 3\varphi < 73/500 < (5/12)^2$. Formula (16) has loss at most $2/21 + \sqrt{457}/42 < 127/210 < c_0$. For B/D, $D_i \ge 10$; the respective losses are $< 1/5 + 2/5$ and $u/6 + \sqrt{u(1 - u)} \le (1 + \sqrt{37})/12 < 3/5$. Thus every foreign isolated cap is strong.

At $Q = 0$ the isolated estimates are also strong, uniformly for $D_i \ge 85/11$. For own 5, the actual old H_4 segment has $\delta_5 = (7 - 3\sqrt{5})/16 < 1/36$ and, by concavity of the old heights, pressure at most γh_i after its correction. Its loss is $< 25/76 + 5/24 < c_0$; here $q_i + \sqrt{D_i - 25/16} > 19/4$

and $\gamma > 4/5$. For own 7, the adjacent negative A_2 core gives loss $< 400/1701 + 9/25 < c_0$, using $.9 < \gamma < 10/11$. For finite own $m \geq 8$, the old singleton projection gives $h_i > \sqrt{D_i - 1} - \tan(\pi/m)$ and loss $< 4/21 + 5/12 < c_0$. For dotted own contact, $c + n_i/\gamma$ gives (16) with $u = \gamma^{-2}$, and the same 127/210 bound. One left H_4 segment suffices: its displacement is $\gamma(3\omega_{i+1}/4 + \omega_{i-3}/4)$ and concavity bounds its scalar product by γh_i.

Here is the complete finite continuation table at a central release. A retained left path A_{i-1} admits endpoint 3, endpoint 4, next-to-end 3, or an admissible E_6, E_7, E_8 position 3. Their Schur residuals are

$$(i+1)/(2i), \quad 1/i, \quad 2/i, \quad 1/4, 1/7, 1/16. \tag{18}$$

For a left 3 at position j from the far end, $\sigma = \gamma + j/(2i)$; for endpoint 4, $\sigma = \gamma + j/(\sqrt{2}i)$. In each case $\cosh L_i = q_i\sigma/\sqrt{\delta - Q}$. The complete right table is

retained right part	finite attachment
$B_s,\ s \geq 4$	long-leaf 3
B_3	long-leaf 3 or short-leaf 3 (F_4)
B_2	3 at either end
$D_s,\ s \geq 5$	long-leaf 3; fork 3 only for $s = 5, 6, 7$
D_4	3 at any leaf
$D_3 = A_3$	middle 3, end $3, 4, 5$.

Two contacts to a retained component create a cycle. A completion joining both long retained pieces has connected rank at least 21, so finite classification forces the corresponding simple endpoint $A/B/D$ bridge. An additional foreign branch at its internal new node is not finite. The exceptional fork extensions in the table are retained and estimated; an E_7 piece is not declared sealed merely because it is exceptional.

Zero own contact needs a separate check. Internal 3, internal 4, right long 3, and two-piece bridges give the usual affine fork, star or cycle; a right D-fork contact gives affine E_8. A zero-own far 3@1 in B/D satisfies $A^2 = Q - 1/2 < 1/(2i) \leq 1/10$; any foreign component then has derivative square at least $1/4 > A^2$, contradicting Lemma 2.2. With $Q = 0$ its height is negative. For A, a zero-own far 3 has height $< 1/(2m)$, a next-to-end 3 has $Q < 2/m \leq 2/5$ and height $Q - 2/N < Q$, and an endpoint 4 has $Q < 1/m \leq 1/5$ and negative height. A positive Q in the first two cases again forces an orthogonal closure. Interior A contacts retain affine E_8 arms $(1, 2, 5)$ or E_7 arms $(1, 3, 3)$. This proves positivity of the own contact without assuming the whole star has no spectators.

For completeness, all other near-position estimates follow directly from (18). Own 3/4/6 is affine except the near 4/simple-3 role. For own $\gamma > 101/125$, near simple 3 has $\delta \leq 3/5$, $\sigma \geq \gamma + 2/5$; the next-to-end row has $\delta = 2/i$, $\sigma \geq \gamma + 1/i$; the E rows have $\delta \leq 1/4$. For B/D, the first row exceeds 100/61 from $i = 8$ onward, and the next-to-end row from $i = 7$ onward. At $i = 5$ use $T(q) < 38/25$ on $[5/2, 10/3]$, and at $i = 6, 7$ use $T(q) < 8/5$ on $[3, 4]$; these are positive concave quadratics after clearing denominators. White 4 is stronger, except far 4/own 3, which retains affine $(3, 4, 3, 3)$. The allowed B near 4/own 3 has $\cosh L_i = (i-1) + i/\sqrt{2}$ and is strong for $i \geq 5$; in D an own 4 with no right contact is already affine on the entire retained right fork. The A near-4 role is kept below.

If a nonisolated far 3 packet has $Q > 0$, the actual data in B/D are

$$\cosh L_i = \frac{2i\gamma + 1}{\sqrt{i + 1 - 2iQ}}, \quad \frac{\cosh L_i}{q_i} > R_0(i) := \frac{(809/500)i + 1}{\sqrt{i(i+2)}}.$$

R_0 increases for $i \geq 5$, with derivative sign $(309/500)i - 1$; T also increases for $q \geq \sqrt{10}$. The three

exact comparison rows are

i	q upper	$R_0(i)$ lower	$T(q)$ upper
5–8	4	1.536	1.530
9–12	5	1.563	1.550
13–29	7.62	1.577	1.577.

The lower squared margins are $.052740, .321913, .546001$; the $(T$ numerator$, T$ denominator$)$ pairs are $(27.4921, 18.0316)$, $(44.2721, 28.6395)$, $(107.2045, 68.003166)$, proving each upper comparison by multiplication. In A, a helper-side parameter $m \geq 5$ instead gives ratio $> (2fm+1)/\sqrt{m(m+2)} \geq 227/(25\sqrt{35}) > 761/500$, $f = 101/125$; the derivative sign is $(2f-1)m - 1 > 0$, with final squared margin 342365. The foreign near-4 role has ratio squared $> 121/35$. Thus every remaining weak packet has $Q = 0$.

4.2 The A_{21} certificate and its plateau transfer

Put $N = r + 1 \in [22, 30]$. Then $D_i = 2i(N-i)/N$, $C_{ij} = 2i(N-j)/N$ for $i < j$, and $\|x_A\|^2 = \frac{1}{2}\sum_{i=0}^{N-1}(h_{i+1} - h_i)^2$, with $h_0 = h_N = 0$ strictly concave. For $5 \leq i \leq N-5$, $85/11 \leq D_i \leq 15$. The preceding inventory leaves a near-4/simple-3 packet or

$$v_i = \gamma_i\omega_i + \omega_{e_i}/2, \quad e_i \in \{1, N-1\}, \quad Q = 0, \quad f := 101/125 < \gamma_i < g := 27/25, \quad A_i > 2. \tag{19}$$

Allowed own contacts are 5, finite $m \geq 7$, or dotted. The dotted upper bound follows because at $\gamma = g$, $\sigma^2/\delta \geq 4g - 2 = 58/25 > (761/500)^2$; the difference numerator is $[2m(g-1) - 1]^2$. The near-4 edge ratio increases with m and at $m = 8$ is $2\sqrt{2}/3 + 7/12 > 761/500$, so $m \geq 8$ is strong. At $m = 7$, $N = 22, 23$ use $D_7 \leq 224/23 < (25/8)^2$ and $T(q) < 3/2$ on $[3, 25/8]$, whose cleared quadratic is $-3400q^2 + 13237q - 7442 > 0$. Every weak near-4 therefore has a mate at index at most 5 for $N = 22, 23$, at most 6 otherwise, or the reversal. That mate is actual by its positive B replacement. Its data are $\delta = 1/j$ and $\cosh L_j = (1/2 + (j-1)/(j\sqrt{2}))\sqrt{jD_j}$. By reversal take $6 \leq j \leq N-4$; then $jD_j \geq 72(N-6)/N \geq 216/5$ and the coefficient is $> 13/12$. Thus $\cosh^2 L_j > 507/10 > 49$, giving normalized height $> \sqrt{3}/2 > 43/50$.

We next exclude adjacent weak literal packets. Write $j = i+1$, $A = \gamma_i$, $B = \gamma_j$, and $R = c_i \cdot c_j$. Weighted AM–GM of their time coefficients gives

$$\begin{aligned}
\text{same helper}: \quad & R \geq -AB(1 - 1/N) \ \pm (A-B)/N \\
& \qquad + (N-1)/(2N) + (N+1)(A/B + B/A)/(4N), \\
\text{outward helpers}: \quad & R \geq 1/2 - AB + (A-1)(B-1)/N, \\
\text{reversed helpers}: \quad & R \geq 1/2 - AB + (A+1)(B+1)/N.
\end{aligned}$$

The first is $> 1 - g^2 > -1/2$; the others are $> 1/2 - g^2 - (1-f)(g-1)/22 > -7/10 > -1/\sqrt{2}$. Hence the native gap gives $R = 0$ for one helper and $R \in \{0, -1/2\}$ for opposite helpers. For outward helpers, $R = 0$ is impossible because $\eta_\ell = \delta_\ell/\sigma_\ell^2 < 4/5$ and

$$(D_i - 4/5)(D_j - 4/5) - C_{ij}^2 = (2/5)C_{ij} - 24/25 + 8/(5N) > 0.$$

For reversed helpers, put $l = i$, $s = N - j$, $l + s = N - 1$. The same determinant, multiplied by N, is

$$4lsA^2B^2 - (l+1)(s+1)(A^2 + B^2) - 2AB - (l+1)A - (s+1)B + (N+2)/4.$$

It increases in $A, B \geq 4/5$: the derivative uses $4lsB^2 - (l+1)(s+1) \geq (39/25)ls - N > 0$. If either contact is at least .9, concavity in l reduces the minimum to $l = 5$ or $s = 5$; at $(4/5, 9/10)$ it is

at least $(509N-9974)/500 \ge 306/125 > 0$. The last possibility has both contacts $\varphi/2$. In raw norm-two coordinates their difference squared is $\varphi^2(N-1)/N < 4$ for the same helper, so their wall product is positive. For opposite helpers, $R = 0$ would have rational coefficient $N(N+6)$ or $-N^2$ in the expansion of $N^2[(v_i \cdot v_j)^2 - A_i^2 A_j^2]$, according to orientation. Neither vanishes. This proves $R = -1/2$ for opposite helpers.

The retained two-cap Schur matrix is $H = I - t\mathbf{1}\mathbf{1}^T$, $t = (l-1)/(2l)$, for one helper; for opposite helpers it has diagonal residuals $\delta_l, \delta_s \in (1/2, 3/5]$ and off-diagonal $-1/2$. It is positive with positive inverse. Put $\lambda = H^{-1}(A_i, A_j)^T > 0$ and $\mathcal{D} = A_i\lambda_i + A_j\lambda_j$. Its actual mixed point has positive displacement $Z = W_i\omega_i + W_j\omega_j$, and

$$Z^2 = \mathcal{D}(1+\mathcal{D}), \qquad x \cdot Z \ge 1 + \mathcal{D} - \sqrt{1+\mathcal{D}}.$$

For one helper, $W > (4/5)l(A_i + A_j)$ and $\mathcal{D} = [l(A_i+A_j)^2 + (A_i - A_j)^2]/2$; thus $\sqrt{1+\mathcal{D}}/W < 7/16$. For opposite helpers, $5/6 < \lambda_i/\lambda_j < 6/5$, $\mathcal{D}/(\lambda_i+\lambda_j)^2 \le 3/55$, $\lambda_i + \lambda_j > 40$, and $W > 32$; the corresponding bound is $< \sqrt{15/176 + 1/1024} < 3/10$. In either case

$$W_i q_i + W_j q_j - \|Z\| < \frac{W}{2(\sqrt{85/11} + \sqrt{80/11})} < 3W/32.$$

The weighted mean loss of two weak indices is therefore below $17/32$ or $63/160$, both below .61. Adjacency is impossible. Every weak near-4 can now be removed: its mate has normalized height $> 43/50$; an opposite mate or a strong anchor among $N-6, N-5$ has height $> 39/50$ and correlation at most $1/3$. For $N \ge 24$ the latter follows from $6/(N-6) \le 1/3$; for $N = 22, 23$ the mate index is at most five and the opposite anchor at least 16, giving correlation at most $30/\sqrt{8160} < 1/3$. Their sum contradicts $(41/25)^2 > 8/3$. Thus all weak central indices have (19) and are nonadjacent.

It remains to exclude weak endpoint 5. We give the complete fixed $N = 22$ certificate before its exact lift. Nonadjacency and concavity give strong 6, a middle floor $h_{11} \ge b$, and a strong 16 or 17, where

$$a = \sqrt{96/11} - .61, \quad b = \sqrt{120/11} - .61, \quad d = \sqrt{85/11} - .61,$$
$$2.344 < a < 2.3443, \qquad 2.6928 < b < 2.693, \qquad 2.169 < d < 2.17.$$

If 16 is strong, the minimum reference x_0 through heights a, b, a at $6, 11, 16$ has $E = x_0^2 = a^2/6 + (b-a)^2/5 > 47/50$ and positive coefficients $c_6 = c_{16} = (11a - 6b)/60 > 4/25$, $c_{11} = (b-a)/5 > 0$. Put $K = 5a/6$, $L = a/12$. For a far helper at 21, each pressure vector in the table has projection $(5\gamma/6)\omega_6 + \omega_{16}/12$, scalar product $\gamma K + L$ with x_0, and height $A^2 = (85/11)\gamma^2 + (5/11)\gamma - 23/44$.

own	pure pressure vector U	target T
5	$\gamma(\omega_4+\omega_6)/2 + \omega_{19}/6$	$H(A, 2/3 - \gamma^2)$
7, 8	$\gamma(\omega_4+\omega_6)/2 + \omega_{20}/4$	$A - \sqrt{\gamma^2 - 3/4}$
≥ 9	$\gamma(\omega_4+\omega_6)/2 + \omega_{21}/2$	$H(A, 1-\gamma^2)$
dotted	$(\gamma - \gamma^{-1})\omega_5 + (\omega_4+\omega_6)/(2\gamma) + \omega_{21}/2$	$A - \sqrt{1-\gamma^{-2}}$

Here $x \cdot U \ge T$, $S = \|U - \operatorname{proj} U\|^2$, and $\Delta = T - \gamma K - L$. The residual variances in row order are $2\gamma^2/3 + 1/12$, $2\gamma^2/3 + 1/6$, $2\gamma^2/3 + 5/12$, and $5\gamma^2/3 + \gamma^{-2} - 19/12$. The exact bounds and chosen positive multipliers are

own	5	7	8	≥ 9	dotted
$\Delta >$	3/10	1/4	9/40	13/50	3/10
$S <$	3/4	3/4	3/4	13/12	5/4
μ	1/5	1/5	1/5	24/125	1/6.

The actual sets are respectively $J = \{5, 21, 20\}$, $\{5, 21\}$, $\{5\}$, and the dotted pair; they use (8) or (7). The following seals cover the full native ranges, not samples. For 5, $\gamma \in (.809, .81)$ gives $A > 2.21$, $\sqrt{2/3 - \gamma^2} < 1/9$, $\gamma K + L < 1.78$. For 7, $\gamma \in (.9, .902)$ gives $A > 2.47$, radical $< .253$, and projection < 1.96. For 8, $\gamma \in (.923, .9239)$ gives $A > 2.5475$, radical $< .322$, projection < 2.0003, so excess $> .2252$. For finite $m \geq 9$, $\gamma > .939$ by the cubic $8\gamma^3 - 6\gamma - 1$; its deficit increases, since $A' > 11/4$, $H_A > 20/23$, $H_\delta < 0$, and $K < 2$. At .939, $A > 2.5917$, $\sqrt{A^2 + \delta} > 2.6144$, $\sqrt{\delta} < .344$, giving $H > 2.29$ and projection < 2.03. The dotted deficit decreases on $(1, 27/25)$ because $A' < 3$, $K > 1.9$, and the radical derivative exceeds $15/8$. At $27/25$, $A > 2.99$, radical $< .38$, projection < 2.306. Its variance increases to a value below $5/4$.

The vector $x_0 + \mu(U - \operatorname{proj} U)$ has all coefficients nonnegative: the largest old-6 removals are $.154, 4/25, 3/20$, and the old-16 removal is at most $1/60$. Its Cauchy margin is $(E + \mu\Delta)^2 - E - \mu^2 S > 0$, increasing in E. The two tight rows have exact lower margins

$$(47/50 + 9/200)^2 - 47/50 - 3/100 = 9/40000,$$
$$(47/50 + (24/125)(13/50))^2 - 47/50 - (24/125)^2(13/12) = 219/39062500.$$

If 17 is strong, the reference through a, b, d has energy $a^2/12 + (b-a)^2/10 + (b-d)^2/12 + d^2/10 > 24/25$. Its old coefficients exceed $4/25$; replace the helper projection by $\omega_{17}/10$. The deficit decreases by less than $1/40$ and variance decreases by $1/60$. The same multipliers still give positive margins (the tight finite-8 margin is already .01). For a helper at 1, reverse the helper term in every vector. The projection joins the ω_6 coefficient; its value on x_0 is unchanged, while A^2 gains $12\gamma/11$. Since the two time coefficients sum to less than $25/4$, $A_{\text{left}} - A_{\text{right}} > 192/1375$. Targets increase by more than .1, using $H_A > 4/5$. The variance gains $\gamma/3$. All finite rows then have $\Delta > 13/40, S < 3/2$ and work with $\mu = 1/6$; dotted rows have $\Delta > 2/5, S < 5/3$ and work with $\mu = 4/25$. Their old-6 removals are at most $11/72, 59/375 < 4/25$. This treats both helpers and both strong anchors and excludes weak 5.

For $N > 22$, express each reference or dual vector by its potential $t_i = W \cdot \omega_i$ and insert $N - 22$ copies of t_{11} between its two old halves. Its norm $\frac{1}{2}\sum(t_{i+1} - t_i)^2$ is unchanged. The only split coefficient is the positive peak at 11, replaced by nonnegative drops at $11, N - 11$. Residual potentials are supported in the unchanged first and last six indices, so all projection coefficients and variances above are exactly preserved. Concavity and nonadjacency give the required middle heights at $11, N - 11$; the old $6, 16/17$ floors can only increase. For the two helpers the time coefficients are

$$A_N^2 = 10\gamma^2 - 1/2 - (10\gamma - 1)^2/(2N), \quad A_N^2 = 10\gamma^2 + 2\gamma - 1/2 - (10\gamma + 1)^2/(2N).$$

They increase with N, as do $H(A, \delta)$ and $A - \sqrt{-\delta}$. Every lifted positive certificate therefore keeps its strict margin. Reversal also excludes weak $N - 5$.

Finally the $N = 22$ full central reference has

$$E_{\text{full}} = (\sqrt{85/11} - .61)^2/5 + \sum_{i=5}^{10}(\sqrt{D_{i+1}} - \sqrt{D_i})^2 > 1007/1000.$$

Lower/upper pairs for $\sqrt{D_5}, \ldots, \sqrt{D_{11}}$ are

$$(2.7797, 2.7799), \quad (2.9541, 2.9543), \quad (3.0895, 3.0897),$$
$$(3.1908, 3.1910), \quad (3.2612, 3.2614), \quad (3.3028, 3.3030),$$
$$(3.3166, 3.3167);$$

squaring them gives lower energy 1.007192468. For $N \geq 23$ lift this same curve with a constant plateau between $11, N - 11$; these two actual heights are at least $\sqrt{11} - .61$ even if weak, since their

strong neighbors have dual square at least $D_{10}(23) = 260/23 > 11$. For $N \geq 23$, only $6, \dots, 10$ and their reversals may now be omitted, with at most three weak indices per flank. For $N = 22$, the weak indices may instead be any nonadjacent subset of $6, \dots, 16$, including the middle index 11; this subset has size at most six. Removing one isolated floor, including that middle floor, loses less than $1/1600$, since $q''(t) = -1/q(t)^3$ and $q > 11/4$ on the original central interval. Segment interiors are disjoint; the surviving potential stays concave with nonnegative duals. Consequently $\|x_A\|^2 > E_{\text{full}} - 6/1600 > 4013/4000 > 1$. This proves the A cap, with all spectator pressure already cleared.

4.3 Classical packets and the two short tails

For B/D put $D_i = 2i$ and $S_i = h_i/\sqrt{2}$. The continuation table in Section 4.1 makes every bulk direction strong except

$$v_i = \gamma_i \omega_i + \omega_1/2, \qquad A_i^2 = 2i\gamma_i^2 + 2\gamma_i - 1/2, \qquad Q = 0, \quad 101/125 < \gamma_i < 6/5. \tag{20}$$

The upper bound follows from $\sigma^2/\delta \geq 4(6/5) - 2 = 14/5 > (100/61)^2$ at $\gamma = 6/5$; the difference numerator is $[2i(\gamma - 1) - 1]^2$. For a right long 3 the data are $\delta = 1/2 - Q$, $\sigma = \gamma + 1/2$; own 0/3 is affine and the next contact gives $\sigma/\sqrt{\delta} \geq 1 + 1/\sqrt{2} > 100/61$. For a right fork 3 in D_s, $s = 5, 6, 7$, $\delta = 1 - s/8 - Q$, $\sigma = \gamma + 1/4$. Own 0/3/4/6 retains affine E_8/cycle/$(3, 3, 4, 3)$ path/$(6, 3)$; the next contact gives ratio at least $(\cos(\pi/5) + 1/4)/\sqrt{3/8} > 1.72 > 100/61$. This includes every possible foreign extension of the finite E_6, E_7, E_8 piece. A two-piece white bridge has residual $1/(2i)$ and numerator at least $\gamma + 1/2 + 1/(2i)$, hence is strong; zero own instead gives an affine cycle.

Adjacent packets (20) are impossible. Weighted AM–GM gives

$$c_i \cdot c_{i+1} \geq 1/2 - ab + (a/b + b/a)/4 \geq 1 - ab > -11/25 > -1/2,$$

so their actual product is zero. Their retained Schur matrix is the same $I - t\mathbf{1}\mathbf{1}^T$, $t = (i-1)/(2i)$, as in Section 4.2. Its positive actual mixed point has $\sqrt{1 + \mathcal{D}}/W < 7/16$ and $W_i\sqrt{2i} + W_{i+1}\sqrt{2i+2} - \|Z\| < W/(4\sqrt{10})$. The derivation uses $A_i > 2$, $\gamma_i > 4/5$ and $i \geq 5$ only: the same explicit inverse and time-coefficient formulas apply. Their sum is below .61, proving nonadjacency.

The index-5 pressure table and its native scalar seals are given in Section 6. Weak 5 forces $x_1 > .5965$, strong 6, and one of $16, 17$ strong. The positive prefix interpolation through $1, 6, 16$ then has energy

$$.5965^2 + (2.01815 - .5965)^2/5 + 1.55051^2/10 = 1.00043812051 > 1. \tag{21}$$

Endpoint 17 increases the last term. Thus 5 is strong using bulk indices alone, before either short-tail estimate below.

In B, the bulk reaches $r - 4$. At $i = r - 3$, the extra right B_3 short-end 3 gives F_4, forbidding every white left or foreign extension. Own 0 is affine; otherwise $\delta = 1/4$, $\sigma \geq 1/2 + 1/(2\sqrt{2})$, strong. At $i = r - 2$, the extra short-end 3 has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/(2\sqrt{2})$. Own 0/4/6 is affine, while own ≥ 5 is strong. Own 3 with $Q > 0$ is strong since $\delta \leq 1/4$. The only new weak role is therefore

$$Q = 0, \qquad v = (\omega_{r-2} + \omega_r)/2. \tag{22}$$

If the short-end contact also joins the left path, the full finite white diagram is B_r and has no foreign extension at c. Its residual $1/(2i)$ and numerator $\geq \gamma + 1/(2i) + 1/(2\sqrt{2})$ are strong even at own 0 for $i \geq 21$. This is a retained zero-own branch, not an assumed affine exclusion. For (22), deleting

$r-1$ and adding the already actual c gives positive A_r, so the same wall actually closes $r-1$. Its exact cosh is

$$C = \frac{(1+\sqrt{2})i+1}{\sqrt{i+3}}, \qquad i = r-2.$$

For $21 \le i \le 23$, $C/\sqrt{2i+2} > 79/50$, $C > 9$, and $\sqrt{2i+2} < 7$. The first comparison reduces to $984i^2 - 19014i - 17498 > 0$, increasing from $i = 21$. Using $\sqrt{C^2-1} > C - 1/C$, its normalized loss is $< 1/(79/50 + 4/63) < .61$. Thus this mate is strong.

In D, the bulk reaches $r-5$. At $i = r-4$ the extra fork 3 has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/4$. Its own 0/3/4/6 is affine as above; $Q > 0$ is strong. At $Q = 0$, own finite ≥ 7 or dotted has ratio > 1.61, above $T(\sqrt{2i}) < 1.577$ for $19 \le i \le 29$. Only own 5 needs the averaged projection below. At $i = r-3$ the full right $D_3 = A_3$ list is middle 3, end 3/4/5. Middle 3 has $\delta = 1/2 - Q$, $\sigma = \gamma + 1/2$ and is affine or strong. End 4 has $\delta = 1/4 - Q$, $\sigma = \gamma + 1/(2\sqrt{2})$; own 0 is affine, otherwise it is strong. End 5 is finite H_4, forbidding further white contacts, but own 0 is retained: already $\delta = 1 - 3\varphi^2/8$ and $\sigma \ge \varphi/4$ give ratio $> 100/61$. End 3 has $\delta = 5/8 - Q$, $\sigma = \gamma + 1/4$. A left bridge is finite A_r, with no further foreign extension; its residual $1/8 + 1/(2i)$ and numerator $\ge \gamma + 1/4 + 1/(2i)$ are strong for $i \ge 21$ unless own 0, which is an affine cycle. Without that bridge, own 0/3/4/6 is affine; $Q > 0$ and $\gamma \ge 1$ are strong. Precisely own 5 or finite own ≥ 7, $\gamma < 1$, $Q = 0$, remains.

For own 5 in either short D layer, put $\gamma = \varphi/2$ and $v = \gamma\omega_i + \omega_f/2$ at the contacted fork. The heights are $A^2 = i(\varphi + 5/8) - 1/2$ or $i(\varphi + 5/8) - 5/8$. Select old paths $i, i-1, i-2$ and $i, i+1, i+2$; at the D_3 layer the last node is the other, uncontacted fork. Each actual H_4 correction is $w_\pm = \gamma(3n_i/2 + n_{\text{side}} + n_{\text{second}}/2)$, with $\delta_5 = 1 - 3\gamma^2/2$. The average $Y = v - (w_+ + w_-)/2$ satisfies

$$x \cdot Y > A - \sqrt{\delta_5}, \quad Y^2 = A^2 + 1 - 15\gamma^2/8 < A^2, \quad Y \cdot u_5 = \sqrt{10}(\gamma + 1/4).$$

The cross product is $w_+ \cdot w_- = 3\gamma^2/4$. For $i \ge 19$, $A > 6$, $\|Y\| > 6.4$, $\sqrt{\delta_5} < 1/7$ (use $\varphi > 809/500$), and $\widehat{Y} \cdot u_5 < 3.3602/6.4 < .55$. Thus $x \cdot \widehat{Y} > 41/42 > .97$; strong 5 gives $x \cdot u_5 > .8$. Their sum contradicts $1.77^2 > 3.1$.

For the finite-≥ 7 D_3 role, $i \ge 21$ and $.9 < \gamma < 1$. Now $A^2 = i(2\gamma^2 + \gamma + 1/8) - 5/8$ and the two actual negative A_2 corrections are $w_\pm = \gamma(4n_i/3 + 2n_{\text{side}}/3)$, with norm residual $-d$, $d = 4\gamma^2/3 - 1 > 0$. Their cross product is $8\gamma^2/9$; the average has

$$x \cdot Y \ge A - \sqrt{d}, \quad Y^2 = A^2 + 1 - 14\gamma^2/9 < A^2, \quad Y \cdot u_5 = \sqrt{10}(\gamma + 1/4).$$

At $i = 21$, $Y^2 - 40(\gamma + 1/4)^2 = 4\gamma^2/9 + \gamma + 1/2 > 0$; larger i only increases it. Hence correlation is $< 1/2$. The ratio d/A^2 increases in γ and decreases in i; its derivative numerator is $[i(4\gamma^2 + 13\gamma + 3) - 5\gamma]/3 > 0$. Its maximum is $1/195$, giving $x \cdot \widehat{Y} > 1 - 1/\sqrt{195} > .928$. Strong 5 gives $x \cdot u_5 > .807$, so their sum contradicts $1.735^2 = 3.010225 > 3$. Every correction is positive and feasible against all unknown walls. Thus the weak inventory (20) extends through $r-3$ in D, with the single B terminal role retained.

We finish the prefix closure explicitly. Let $f_j = \sqrt{j} - .61/\sqrt{2}$ and $E_M = f_5^2/5 + \sum_{j=5}^{M-1}(\sqrt{j+1} - \sqrt{j})^2$. The concave reference has nonnegative duals. The exact seals are $E_{21} > 1.009$, $E_{20} > .99752$, $f_5/5 < .361$: $f_5^2/5 > .6514$ and the increment lower bounds for $j = 5, \dots, 20$, in units 10^{-5}, are

$$4545, 3846, 3333, 2941, 2631, 2380, 2173, 2000, 1851, 1724, 1612, 1515, 1428, 1351, 1282, 1219.$$

They follow from $(\sqrt{j+1} - \sqrt{j})^2 > 1/(4j+2)$. One isolated missing floor loses $< 1/3200$ by $|(\sqrt{t})''| < 1/40$; two consecutive missing floors lose $< 1/800$, by the variance of their three segment slopes. Disjoint segment interiors make these losses additive.

For a far weak packet at any $5 \le i \le 21$, the actual pressure table in Section 6, with $q = \sqrt{2i}$, gives $x_1 > \sqrt{2}[T - \gamma(q - .61)] > .424$. For finite own $5, 7, 8, \ge 9$, use $A - \gamma q > .079$ and the four bracket lower bounds $.61(.809) + .079 - .111$, $.61(.900) + .079 - .253$, $.61(.923) + .079 - .323$, $.61(.939) + .079 - .344$; each times $7/5$ exceeds $.424$. For dotted $\gamma \in (1, 6/5)$ the bracket decreases in both γ, q: $A' - q < (q^2 + 2)/(4q^3) < .1$, while the radical derivative exceeds 1. Its endpoint bound is

$$\sqrt{2}[\sqrt{3119/50} - (6/5)\sqrt{42} + .732 - \sqrt{11}/6] > .424,$$

sealed by $7.8981 < \sqrt{3119/50}$, $\sqrt{42} < 6.480741$, $\sqrt{11}/6 < .552771$, and $\sqrt{2} > 1.4142$.

In D_r, $r \ge 24$, index 21 is now in range. If strong, $E_{21} - 8/3200 > 1$. Otherwise 20 is strong and $x_1 > .424$; the reference through 20, with this first-coordinate constraint, has energy

$$E_{20} + \frac{5}{4}(.424 - .361)^2 - 7/3200 > .99752 + .00496125 - .0021875 = 1.00029375 > 1. \tag{23}$$

Its slopes still decrease, so all dual coefficients are nonnegative. For B_r, $r \ge 26$, one of $21, 22$ is strong and the same E_{21} argument applies. At $r = 25$ it still applies through $r - 3$. At $r = 24$, choose strong 21 or 22, or the strong mate 23 of (22). At $r = 23$, a terminal role at 21 likewise supplies strong 22; a far weak 21 instead uses (23). In these terminal cases there is at most one two-hole cluster; the excessive loss $9/3200 + 1/800 < .009$ still leaves energy above one. This proves both classical caps and completes Proposition 4.2.

5 Coupled pressure in large A components

The crucial feature of this section is that a foot at one end is tested against *actual* opposites at the other end. A collection of separate edge estimates would lose that coupling.

Theorem 5.1 (Uniform actual A pressure)**.** *With arbitrary finite spectators,*

$$\|x_{A_r}\|^2 > \frac{1}{2} \quad (r \ge 7), \qquad \|x_{A_r}\|^2 > \frac{5}{7} \quad (r \ge 17).$$

We first prove the two rank-uniform steps, and then give the rank-seven base in Subsection 5.4. Put $N = r + 1$, and realize

$$n_i = (e_i - e_{i+1})/\sqrt{2}, \quad C_{ij} = \frac{2\min(i, j)(N - \max(i, j))}{N}, \quad h_i = x_A \cdot u_i.$$

The root space is the sum-zero subspace of $\mathbb{R}^N$. All its inequalities below use the actual projection x_A.

5.1 A forcing window and the complete common-pair inventory

Suppose first that $N \ge 9$ and $\|x_A\|^2 \le 1/2$. Consider the four pairs $1/2, 3/4, (r-3)/(r-2), (r-1)/r$. If their respective opposites are distinct, the outer pair sums exceed 1, and the inner sums exceed $19/16$. These follow from (4), concavity of $\log F$, and the minimum correlations $\sqrt{3/7}$, $\sqrt{3/5}$ at $N = 8$. For the inner comparison,

$$F(19/32)^2 = \frac{1478656}{1918225} < \frac{387}{500} < \sqrt{3/5}.$$

For the outer one, $F(1/2)^2 = 16/25 < \sqrt{3/7}$. Set

$$L = 2u_1 + 2u_2 + u_3 + u_4, \qquad R = 2u_r + 2u_{r-1} + u_{r-2} + u_{r-3}.$$

The correlations on the left increase to $\sqrt{i/j}$, whence

$$L^2, R^2 < 12 + 6\sqrt{2} + 4/\sqrt{3} + 4\sqrt{2/3} + \sqrt{3} < 28.$$

The last bound follows, for example, by the positive rational upper bound 2351/84. For $a = (2, 2, 1, 1)$, direct coordinates give

$$L \cdot R = \left(\sum_{i=1}^{4} a_i \sqrt{\frac{i}{N-i}} \right)^2 .$$

This decreases in N; at $N = 9$ its square root is $\sqrt{2} + 2\sqrt{2/7} + 2/\sqrt{5} < 17/5$. Thus $(L + R)^2 < 80$, whereas distinctness of all four pairs gives $x_A \cdot (L + R) > 51/8$. The Cauchy gap is $41/64 > 0$. After reversing the numbering, a common pair is therefore $1/2$ or $3/4$. This is a relabeling, not a symmetry assumption on P.

Here is the finite inventory used at both targets in Theorem 5.1. It is useful to state it before making any energy-dependent discard. Write Q for the total spectator cost. Every retained finite diagram is a forest, so there is at most one contact on each retained own path. The foreign costs satisfy (6), are at least $1/4$, and have the gap $(1/4, 1/3)$.

For a common $3/4$ opposite, its own contact positions are in $\{1, 2, 3\}$ and $\{4, \dots, r\}$. A double profile away from positions $3, 4$ survives one deletion as a connected rank-r finite diagram. Its two new bonds must be 3: a weighted bond would be internal in a B path, and D, E_8 are simply laced. The two 3's instead give an affine cycle in the full diagram. This also covers the rank-eight E_8 boundary. At positions $3, 4$, the labels are

$$p \in \{3, 4, 5\}, \qquad q \in \{3, 4\}.$$

The pair $3, 3$ is affine. The two smallest remaining assignments have actual endpoint cosines

$$\sqrt{\frac{3}{2N}}\big(N - 3 + \sqrt{2}(N - 4)\big), \qquad (3\sqrt{2} + 4)\sqrt{\frac{N-4}{2N}}, \tag{24}$$

both greater than 3 at $N = 9$ and increasing thereafter. Increasing a contact or Q strengthens the appropriate endpoint.

A single 3 at positions $3, \dots, r - 2$ contains an affine E_8 tree with arms $1, 2, 5$, or an affine E_7 tree with arms $1, 3, 3$, obtained by truncation if needed. The other single-3 possibilities satisfy

position	Q bound	A^2
1	$Q < 5/8$	$Q - (N + 1)/(2N) < 1/8$
2	$Q < 1/2$	$Q - 2/N$
$r - 1$	$Q < 2/(N - 3)$	$Q - 2/N$
r	$Q < 1/2 + 1/(2(N - 3))$	$Q - (N + 1)/(2N) < 3/(2N(N - 3))$.

The end heights are below $1/4$; the middle rows allow at most one foreign component and its cost exceeds A^2. All contradict (6). A single 4 is an affine $3, 3, 4$ star internally; at either end its white condition gives $Q < 1/4$, so $Q = 0$ and its height is negative. A single 5 can occur only at 1 or 3, has $Q = 0$ from its H_4 cost $3\varphi^2/8 > 3/4$, and at 3 closes direction 4 with

$$\cosh L_4 = 3\varphi^3 \sqrt{(N - 4)/N}. \tag{25}$$

At 1 it gives foot F below. A zero own contact cannot close an own edge. This completes the common $3/4$ list.

For a common $1/2$ opposite there is possibly a contact at 1 and at most one on $2,\dots,r$. For double positions $1,j$, $j\ge 3$, deletion 2 gives a full connected finite path or branch. A branch has only 3's; a path can have a 4 only at its terminal bond $1-c$, with $j=3$ or r. All 3's give an affine cycle. The profile $4@1+3@r$ contains the affine F_4 path $2,1,c,r,r-1$. The remaining $4@1+3@3$ has $Q<1/(2(r-1))$, hence $Q=0$, and gives foot H. Adjacent contacts $1,2$ have an arbitrary finite native first label and second label 3 or 4; $3,3$ is affine.

For single 3, the same internal affine exclusions apply. At $1,r,r-1$, respectively, the bounds are

$$Q<3/4,\quad Q<1/2+1/(2r),\quad Q<2/r.$$

The first two give height below $1/4$, and the last leaves no foreign cost and negative height. At 2, $Q<1/2+1/(2r)$ allows at most two foreign components. One exceeds $A^2=Q-2/N$. With two, both contacts are 3; c,n_2,n_1,n_3 and their two foreign neighbors give the exact affine two-fork diagram. A single 4 is affine internally; at 1, $Q<1/2$ allows at most one cost greater than $A^2=Q-1/N$; at r, $Q<1/r$ leaves negative height. A single label at least 5 is possible only at 1. Label 6 gives the affine $3,6$ path. Labels at least 7 have $Q<1-\beta_1^2<1/4$, hence $Q=0$. Label 5 with $Q=0$ gives F; with $Q>0$ it remains in L. Thus the remaining end profiles are

$$\text{L}:\ \text{adjacent, not } (3,3);\ \text{isolated native } \ge 7;\ \text{isolated 5 with } Q>0,\qquad \text{H},\text{F}.$$

No foreign branch has been suppressed by assuming a full star finite.

5.2 Three feasible feet prove the half bound

For H and F take the following positive actual combinations:

	ξ	$k=-\xi^2$
H	$c+(\sqrt{2}-\frac{1}{2})n_1+(\sqrt{2}-1)n_2+\frac{\sqrt{2}}{2}n_3$	$\sqrt{2}-\frac{5}{4}$
F	$c+\sum_{j=1}^{4}((j+1)\varphi-2j)n_j$	$34\varphi-55.$

For H the products with c,n_1,n_3 vanish and the n_2 product is $(\sqrt{2}-3)/4<0$. For F the products with c,n_1,n_2,n_3 vanish and the n_4 product is $3\varphi-5<0$. Every other actual inequality follows from positivity. Both have $Q=0$, future coefficient $A>0$, and pure A-spatial part, with

$$A_{\text{H}}^2=\frac{3}{2}+\sqrt{2}-\frac{11/2+3\sqrt{2}}{N},\qquad A_{\text{F}}^2=\frac{\varphi^2(N-1)}{2N}-1.$$

At $N=9$,

$$A_{\text{H}}^2-9k_{\text{H}}=\frac{437-300\sqrt{2}}{36}>0,\qquad A_{\text{F}}^2-9k_{\text{F}}=\frac{50(89-55\varphi)}{9}>0.$$

The time norms increase with N. Equation (9) therefore excludes these profiles under energy at most $1/2$. The central doubles and (25) already have $\cosh L>3$, so are excluded by (2).

For L, first clear all foreign components positively to obtain $c_F=Ap-v_A$, $c_F^2=1-Q$. Put

$$b=\beta_1,\ d=\beta_2,\ B=b+2d,\quad h=\frac{4}{3}(b^2+bd+d^2)+Q-1,\quad z_0=h/B^2,$$

$$\lambda_1=(4b+2d)/3-h/B,\qquad \lambda_2=2B/3-2h/B,\qquad \xi=c_F+\lambda_1n_1+\lambda_2n_2=Ap-U.$$

The endpoint condition gives

$$B\lambda_1 = 2bd + 1 - Q > 0, \quad B\lambda_2 = 2(1 - b^2 - Q) > 0,$$

and direct multiplication gives

$$\xi \cdot c = \xi \cdot n_1 = \xi \cdot S = 0, \quad \xi \cdot n_2 = -3h/(2B) < 0, \quad k = -\xi^2 = B^2(z_0 - 3z_0^2).$$

Here S denotes all cleared spectator roots. The complete L range is

$$\frac{8}{81} < z_0 < \frac{1}{3}. \tag{26}$$

Indeed adjacent contacts with one label at least 4 give $z_0 > 1/10$; isolated labels at least 7 have $b > 9/10$, so $z_0 = 4/3 - b^{-2} > 8/81$; isolated 5 with $Q \geq 1/4$ gives $z_0 \geq 3\varphi - 14/3 > 2/15$. The upper bound is precisely $1 - b^2 - Q > 0$. Thus ξ is actual, future timelike and has no foreign spatial part.

For $a_N = 2/3 - 2/N$, its full vector data are

$$A^2 = B^2(a_N + z_0), \quad U^2 = B^2(a_N + 3z_0^2), \quad U_{\text{raw}} = \frac{B}{\sqrt{2}}(2/3 + z_0, 2/3 + z_0, 2/3 - 2z_0, 0, \ldots, 0). \tag{27}$$

Subtracting $\sqrt{2}B/N$ from every coordinate centers the raw vector. Its normalized actual demand $g = (A - \sqrt{k})/\sqrt{U^2}$ satisfies $g^2 > 9/20$. At $a_N = 4/9$ the equivalent quadratic is $2523z_0^2 - 720z_0 + 484/9 > 0$, of negative discriminant $-72976/3$; increasing N strengthens it.

Let $W = u_{r-1} + u_r$. Direct coordinates give

$$\rho = \frac{U \cdot W}{\sqrt{U^2}} = \frac{2/\sqrt{N(N-2)} + \sqrt{2/(N(N-1))}}{\sqrt{a_N + 3z_0^2}}. \tag{28}$$

The numerator decreases and denominator increases with N. At $N = 9, z_0 > 8/81$, its squared upper seal is $48114/129500 < 3721/10000$, so $\rho < .61$. If the far opposites are distinct, $x_A \cdot W > 1$. Perpendicular Cauchy instead gives

$$x_A \cdot W < \frac{.61}{\sqrt{2}} + \sqrt{(1/2 - 9/20)4} < .89 < 1.$$

If they are common, their inventory supplies another L foot: H and F have already been excluded. Normal reversal changes the raw sum from $\sqrt{2}B$ to $-\sqrt{2}B'$. The disjoint raw supports therefore give the *positive* centered product

$$U \cdot U' = 2BB'/N, \qquad \operatorname{corr}(U, U') < \frac{2/N}{a_N} = \frac{3}{N-3} \leq \frac{1}{2}.$$

The two actual demands force energy greater than $4(9/20)/3 = 3/5$. This proves the half bound.

5.3 A fourth foot raises the cost to $5/7$

Now $N \geq 18$ and suppose $\|x_A\|^2 \leq 5/7$. For distinct opposites the outer pair sums exceed $53/50$, and the inner sums exceed $13/10$. Exact calibrations are

$$F(53/100)^2 = \frac{112360000}{164070481} < \frac{137}{200} < \sqrt{8/17}, \quad F(13/20)^2 = \frac{270400}{323761} < \frac{209}{250} < \sqrt{7/10}.$$

At $N = 18$ the square root of $L \cdot R$ is

$$\frac{2}{\sqrt{17}} + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{5}} + \frac{2}{\sqrt{14}} < \frac{1089}{500}.$$

Consequently $(L+R)^2 < 6549/100$, while four distinct pairs give $x_A \cdot (L+R) > 171/25$. The exact gap

$$(171/25)^2 - \frac{5}{7}(6549/100) = \frac{123}{17500} > 0$$

again forces an end or central common pair.

The complete inventory above is threshold-independent. At the new target the relevant edge threshold is $\cosh L = 6$, so the old central discards must be revisited. For adjacent $p@3, q@4$, $q \geq 4$ gives the first expression in (24), now greater than 6. For $p = 5, q = 3$, the direction-four cosine is

$$(3\varphi + 4)\varphi^2\sqrt{(N-4)/N} > 6.$$

The isolated 5@3 cosine in (25) is also greater than 6. These bounds hold already at $N = 18$ and increase with N. The remaining $p = 4, q = 3$ has $Q = 0$ and requires a new actual foot:

$$\mathrm{C}: \quad \xi = c + n_3 + (2 - \sqrt{2})n_4, \quad k = 3\sqrt{2} - 4. \tag{29}$$

Its products with c, n_3 vanish and its n_4 product is $1 - \sqrt{2} < 0$. Its other products follow from positivity. It is future, with pure A-spatial part, and

$$A^2 = 4 + 3\sqrt{2} - \frac{17 + 12\sqrt{2}}{N}, \quad U^2 = 8 - \frac{17 + 12\sqrt{2}}{N}, \quad B = 3\sqrt{2}/2 + 2.$$

Here and below B is the raw coordinate sum divided by $\sqrt{2}$. A central common pair now supplies C or F; an end common pair supplies L, H or F. None is discarded using an old threshold.

All four feet have positive raw sums, support in at most the first five coordinates, and the following uniform bounds:

foot	raw support	g^2	(U^2/B^2)
L	$1, \ldots, 3$	$> 49/100$	$> 5/9$
H	$1, \ldots, 4$	$> 29/50$	$> 4/9$
F	$1, \ldots, 5$	$> 3/5$	$> 1/3$
C	$1, \ldots, 5$	$> 13/20$	$> 1/3$.

For completeness the scalar seals at $N = 18$ are as follows. For L the demand inequality is $66603z_0^2 - 19600z_0 + 1445 > 0$, with discriminant -805340. For H,

$$A^2 = (43 + 30\sqrt{2})/36 > 237/100, \quad k < 33/200, \quad U^2/B^2 = \frac{88 - 6\sqrt{2}}{9(11 + 6\sqrt{2})} > 4/9.$$

Their ratio exceeds $158/11 > (79/21)^2$. For F,

$$A^2 = (17\varphi - 19)/36 > 59/250, \quad k < 167/12500, \quad U^2/B^2 = \frac{1961 - 1207\varphi}{9(\varphi + 1)} > 1/3,$$

and the ratio exceeds $2950/167 > 16$. For C,

$$A^2 = (55 + 42\sqrt{2})/18 > 569/90, \quad k < 1/4, \quad U^2/B^2 = \frac{127 - 12\sqrt{2}}{9(17 + 12\sqrt{2})} > 1/3;$$

the ratio exceeds $1138/45 > (33/7)^2$. All comparisons follow by positive rational squaring, using $1.414 < \sqrt{2} < 1.415$ and $1.618 < \varphi < 1.61804$; for the F density it suffices that $\varphi < 1.6181$. The time norms and densities increase with N, while k and B are fixed. For L the estimates hold on its entire domain.

For the far pair W, $W^2 < 7/2$. The numerator in (28) is below

$$\sqrt{2}/12 + 1/\sqrt{153} < 159/800.$$

The density bounds give $\rho_H < 3/10$ and $\rho_F, \rho_C < 7/20$. For L keep (26):

$$\rho_L^2 < \frac{(159/800)^2}{5/9 + 64/2187} = \frac{55289547}{818560000} < \frac{169}{2500}.$$

If the far opposites are distinct, their sum must exceed $53/50$. For L the Cauchy bound decreases with its normalized coordinate $v \geq 7/10$ and increases with $\rho \leq 13/50$. Thus

$$x_A \cdot W < \frac{91}{500} + \sqrt{\frac{1347217}{1750000}} < \frac{91 + 439}{500} = \frac{53}{50};$$

the last squared integer margin is $1830 > 0$. For H, F, C, respectively, the simpler perpendicular bounds are

$$\tfrac{3}{10}\sqrt{5/7} + \sqrt{47/100} < .95, \quad \tfrac{7}{20}\sqrt{5/7} + \sqrt{2/5} < .94, \quad \tfrac{7}{20}\sqrt{5/7} + \sqrt{9/40} < .78.$$

All contradict distinctness.

Finally suppose the far opposites are common. Their inventory gives an L, H or F foot at that end. Both raw supports are disjoint and every density is greater than $1/3$. The positive centered product $2BB'/N$ therefore gives correlation below $1/3$. Each normalized demand exceeds $7/10$. Their sum forces

$$\|x_A\|^2 > \frac{(7/5)^2}{2 + 2/3} = \frac{147}{200} > \frac{5}{7}.$$

This proves the half bound for $r \geq 8$ and the $5/7$ bound for $r \geq 17$. It remains to supply the rank-seven base.

5.4 The rank-seven base

Set $N = 8$ and suppose $\|x_A\|^2 \leq 1/2$. The correlations of pairs $1/2, 6/7$ are $\sqrt{3/7}$ and those of $3/4, 4/5$ are $\sqrt{3/5}$. The pair-sum calibrations already proved therefore apply. If all four pairs have distinct opposites, put

$$w = 2u_1 + 2u_2 + u_3 + 2u_4 + u_5 + 2u_6 + 2u_7.$$

Then $x_A \cdot w > 51/8$, whereas direct Gram summation gives

$$\begin{aligned} w^2 = {} & \frac{64}{7} + \frac{32}{3} + \frac{16}{5} + 4 \\ & + \frac{64}{\sqrt{21}} + \frac{64}{\sqrt{105}} + \frac{16}{\sqrt{7}} + \frac{64}{3\sqrt{5}} \\ & + \frac{16}{\sqrt{3}} + 8\sqrt{3/5} < \frac{1208}{15}. \end{aligned}$$

The first four terms sum to less than 28, and the last six are respectively less than 14, $32/5$, $32/5$, 10, $28/3$, $32/5$; each comparison follows by squaring positive rational bounds. Cauchy is impossible since

$$(51/8)^2 - (1208/15)/2 = 359/960 > 0.$$

Reversing the numbering if necessary, a common pair is $1/2$ or $3/4$.

The central pair. For a common $3/4$ opposite there is at most one contact on each of the paths $1,2,3$ and $4,5,6,7$. Native bonds have labels $3,4,5$: a larger label with a retained old bond is not finite. For two contacts outside positions $3,4$, the following finite white tests exhaust the position pairs:

$(1,4),(1,7),(2,4),(2,7),(3,5),(3,7)$	a rank-seven path after deleting 3 or 4
$(1,5),(1,6),(2,5),(2,6)$	D_7 or E_7 after deleting 3
$(3,6)$	D_7 after deleting 4.

Both new bonds must be 3 in these rows; their cycle in the full diagram is affine. At positions $3,4$, the $3,3$ pair is affine; every other assignment gives an endpoint cosine greater than 3, already for the minimal $3,4$ and $4,3$ assignments in (24) at $N=8$. It therefore forces energy greater than $1/2$.

For a single 3-contact, the complete own costs and sufficient white bounds are

j	1	2	3	4	5	6	7
$\beta^t C\beta$	7/16	3/4	15/16	1	15/16	3/4	7/16
$Q <$	5/8	1/2	5/8	3/5	2/5	2/5	3/5.

Here $A^2=\beta^tC\beta+Q-1$. One foreign component is impossible by $q<A^2$ in (6). Every row except $j=3,4$ allows at most one, or has $A^2<1/4$. In rows $3,4$, two foreign components must both start with a 3-contact; the new wall, the contacted old node, its two old neighbors and the two foreign neighbors give the exact affine two-fork D_5 diagram. The case $Q=0$ has nonpositive height.

A single 4 is an affine $3,3,4$ star internally. At an end $Q<1/4$, so $Q=0$ and the height is negative. A single 5 is possible only at positions 1 or 3, and its retained H_4 cost $3\varphi^2/8>3/4$ forces $Q=0$. Position 3 gives direction-four cosine greater than 3. Position 1 gives foot F above. At $N=8$ its time norm satisfies

$$A_F^2=(7\varphi-9)/16, \qquad A_F^2-9k_F=(7911-4889\varphi)/16>0.$$

For example $\varphi<16181/10000$, proved by positive squaring, gives $7911-4889\varphi>1091/10000$. Thus the actual-point inequality (9) excludes this last row. The central common-pair case is complete.

The end pair. For common $1/2$, a double profile has positions $1,j$. For $j=2$ its second label is 3 or 4 and its first label is any finite native label. At $j=3,7$ the labels are 3 at j and 3 or 4 at 1. At $j=4,6$, deletion 2 gives E_7 and both labels are 3. At $j=5$, deletion 2 already contains affine E_6. Two 3-labels always give an affine cycle. The $4@1+3@7$ row contains the affine F_4 path $2,1,c,7,6$; the $4@1+3@3$ row has $Q=0$ and is foot H. The displayed H formula gives $A_H^2-9k_H=(193-134\sqrt{2})/16>0$ at $N=8$.

For a single 3, the own costs are the same seven entries above, while the sufficient Q bounds at positions $1,\dots,7$ are

$$3/4,\quad 4/7,\quad 2/7,\quad 1/7,\quad 1/7,\quad 2/7,\quad 4/7.$$

All except position 2 have $A^2<1/4$. At position 2, one foreign component contradicts $q<A^2$; with two, each exceeds $1/4$, so each is at least $1/3$, contrary to $Q<4/7$. A single 4 is affine internally. At 1, $Q<1/2$ permits at most one foreign component, whose cost exceeds $A^2=Q-1/8$; at 7, $Q<1/7$ leaves none and gives negative height.

A single label at least 5 is possible only at 1. Label 6 contains the affine $3,6$ path. A label at least 7 forces $Q<1-\beta_1^2<1/4$, hence $Q=0$. Label 5 with $Q=0$ was excluded by F. Thus the only remaining profiles are exactly L: adjacent contacts other than $3,3$, isolated label at least 7 with $Q=0$, or isolated 5 with $Q>0$.

The far-end comparison. The positive spectator clearing and actual L-foot formulas (26)–(27) remain valid at $N = 8$, now with $a_N = 5/12$. The lower bound $z_0 > 8/81$ follows as before: adjacent contacts give $z_0 > 1/10$, an isolated label at least 7 has $\beta_1 > 9/10$, and isolated 5 with $Q > 0$ has $z_0 \geq 3\varphi - 14/3 > 2/15$. For the label-seven bound, $8t^3 - 4t^2 - 4t + 1$ is $-1/125$ at $9/10$ and strictly increasing from $\sqrt{3}/2$ to 1.

The actual demand now satisfies $g^2 > 11/25$: equivalently $A^2/k > 324/49$, whose cleared difference is

$$972z_0^2 - 275z_0 + 245/12 > 0.$$

Its discriminant is -3755. With $W = u_6 + u_7$, direct coordinates give

$$0 < \rho^2 = \frac{5/42 + 1/(2\sqrt{21})}{5/12 + 3z_0^2} < \frac{603612}{1170300} < \frac{324}{625}, \qquad W^2 = 2 + 2\sqrt{3/7} < 10/3.$$

Here the numerator is less than $23/100$ and $z_0^2 > 64/6561$ gives the stated rational seal. If the far opposites are distinct, their pair law gives $x_A \cdot W > 1$. Perpendicular Cauchy instead gives

$$x_A \cdot W < \frac{18}{25\sqrt{2}} + \sqrt{(1/2 - 11/25)10/3} = \frac{18}{25\sqrt{2}} + \frac{1}{\sqrt{5}} < \frac{24}{25}.$$

If they are common, the reversed end inventory supplies another L foot. Normal reversal is $e_j \mapsto -e_{9-j}$, so its raw sum is $-\sqrt{2}B'$ on the disjoint final three coordinates. Consequently $U \cdot U' = BB'/4 > 0$, and their normalized correlation is less than $3/5$. The sum of these two normalized directions has square less than $16/5$ and actual scalar product greater than $2\sqrt{11/25}$. Cauchy forces

$$\|x_A\|^2 > \frac{44/25}{16/5} = \frac{11}{20} > \frac{1}{2}.$$

Both far-end alternatives are impossible. This proves the rank-seven base and completes Theorem 5.1.

5.5 The dimension-twenty A frontier

Proposition 5.2. *An actual A_{20} component has energy greater than $11/14$, with arbitrary finite spectators.*

Proof. Put $N = 21$ and suppose the energy is at most $11/14$. Keep the outer and inner pair-sum calibrations $53/50, 13/10$, which are independent of this energy target. Flip the forcing weights to

$$L = u_1 + u_2 + 2u_3 + 2u_4, \qquad R = 2u_{17} + 2u_{18} + u_{19} + u_{20}.$$

The six off-diagonal terms of L^2 have respective upper bounds

$$\begin{array}{lll} 2\sqrt{19/40} < 69/50, & 4\sqrt{3/10} < 11/5, & 4\sqrt{17/80} < 37/20, \\ 4\sqrt{12/19} < 16/5, & 4\sqrt{17/38} < 67/25, & 8\sqrt{17/24} < 337/50. \end{array}$$

Adding the diagonal gives $L^2, R^2 < 561/20$. Moreover

$$\sqrt{L \cdot R} = 1/\sqrt{20} + \sqrt{2/19} + 2/\sqrt{6} + 4/\sqrt{17} < 9/40 + 13/40 + 41/50 + 49/50 = 47/20.$$

Thus $(L + R)^2 < 13429/200 < 68$. Four distinct pairs would give $x_A \cdot (L + R) > 183/25$, with Cauchy gap

$$(183/25)^2 - (11/14)68 = 673/4375 > 0.$$

A common 1/2 or 3/4 pair is forced after relabeling.

At this target the edge discard is $\cosh L > 25/3$. For the central adjacent $q \geq 4$ row the minimum is

$$(18 + 17\sqrt{2})/\sqrt{14} > 836/75 > 25/3.$$

For $p = 5, q = 3$ it is $(3\varphi + 4)\varphi^2\sqrt{17/21} > 4576/225 > 25/3$. The isolated 5@3 value is $3\varphi^3\sqrt{17/21} > 32/3 > 25/3$. These rational seals use $\sqrt{2} > 7/5, \sqrt{14} < 15/4$, $\varphi > 8/5, \varphi^2 > 13/5, \varphi^3 > 4, \sqrt{17/21} > 8/9$. The remaining C, H and F profiles are all retained. The finite and foreign inventory above has no energy-dependent discard left unexamined. Every common pair supplies L, H, F or C.

If either far pair is common, its reflected foot has disjoint raw support. The positive centered product is $2BB'/21$. The following three estimates cover all four-foot pairings:

types	correlation upper bound	energy forced above
L/L	6/35	343/410
L/non-L	1/4	5329/6250
non-L/non-L	2/7	203/225.

Indeed L has demand $> 7/10$ and density $> 5/9$; every non-L has demand $> 19/25$, since $(19/25)^2 < 29/50$, and density $> 1/3$. For the middle row the squared correlation is $< 12/245 < 1/16$. Sum-direction Cauchy gives every displayed cost, each above 11/14. This excludes a common far pair.

If both far pairs are distinct, $x_A \cdot R > 183/50$. For any left foot its raw support is before index17 and

$$U \cdot R/B = K = 2\sqrt{8/357} + 2/\sqrt{63} + 2/\sqrt{399} + 1/\sqrt{210} < 3/10 + 13/50 + 11/100 + 7/100 = 37/50.$$

Write $\rho = (U \cdot R)/\sqrt{U^2} = K/\sqrt{d_U}$. This is a coefficient against a unit foot, not a unit-unit correlation, and $R^2 < 561/20$. For L, $\rho^2 < 12321/12500 < 1$ and its actual normalized coordinate $v > 7/10$. Perpendicular Cauchy gives

$$x_A \cdot R \leq \rho v + \sqrt{(11/14 - v^2)(561/20 - \rho^2)}.$$

It decreases with $v \geq .7$ and increases with $0 \leq \rho \leq 1$: both derivative signs reduce to $v^2(561/20) > (11/14)\rho^2$. Therefore

$$x_A \cdot R < .7 + \sqrt{111987/14000} < .7 + 2.9 = 18/5 < 183/50.$$

For every non-L, $\rho < (3/4)\sqrt{3} < 4/3$ and $v^2 > 29/50$. The separate perpendicular bound instead gives

$$x_A \cdot R < \frac{4}{3}\sqrt{11/14} + \sqrt{(11/14 - 29/50)(561/20)} < \frac{6}{5} + \frac{121}{50} = \frac{181}{50} < \frac{183}{50}.$$

The two squared rational margins are 68/1575 and 1507/17500. Both far-distinct alternatives are impossible, proving the proposition. □

6 One generic classical prefix estimate

The next estimate keeps the opposite-wall pressure even when the first reference index is weak. It avoids proving that index strong by a contradiction using the total norm.

Theorem 6.1 (Actual prefix cost)**.** *Fix $6 \le M \le 16$. A whole B_r component with $r \ge M+5$, or D_r with $r \ge M+6$, has*

$$\sum_{j=1}^{M+1} x_j^2 > E_M - \frac{H_M}{3200}, \qquad E_M = \frac{f_5^2}{5} + \sum_{i=5}^{M-1} d_i^2, \quad H_M = \left\lfloor \frac{M-5}{2} \right\rfloor,$$

where

$$f_i = \sqrt{i} - \frac{61}{100\sqrt{2}}, \qquad d_i = \sqrt{i+1} - \sqrt{i}.$$

All coordinates belong to the same actual component projection. Spectators are arbitrary.

In the relevant long-root window,

$$\omega_i = \sqrt{2}(e_1 + \cdots + e_i), \quad C_{ij} = 2\min(i,j), \quad S_i = \sum_{j=1}^{i} x_j.$$

Call index i strong if $S_i > f_i$, otherwise weak. The finite continuation needed here is the bulk calculation in Section 4.1, not the final component exclusions. Its complete weak inventory on $5 \le i \le M+1$ is

$$v = \gamma\omega_i + \omega_1/2, \quad A^2 = 2i\gamma^2 + 2\gamma - \tfrac{1}{2}, \quad Q = 0, \quad \frac{101}{125} < \gamma < \frac{6}{5}, \tag{30}$$

with own finite label 5, at least 7, or a dotted contact. This is the complete bulk inventory proved in Section 4.1; it is not inferred solely from the prefix matrix.

We spell out its rank and actualness boundary. Deletion of i leaves a left A_{i-1} of rank at least 4 and a right B of rank at least 4, or D of rank at least 5. A full own white bridge has rank greater than 8; its finite type is B or D, so an additional foreign branch is sealed. The right-fork D-extensions to E_6, E_7, E_8 are retained and estimated as strong. Foreign isolated clearing and the zero-own-contact repair keep every spectator. The foreign-far comparison needs only $i \le 17$, within its proved range $5 \le i \le 29$. Neither short D_4/D_3 tail nor a rank cutoff is used here.

For adjacent weak indices $i, i+1$, with parameters a, b from (30), their actual wall product satisfies

$$c_i \cdot c_{i+1} \ge \frac{1}{2} - ab + \frac{1}{4}(a/b + b/a) \ge 1 - ab > -\frac{11}{25} > -\frac{1}{2}.$$

It must therefore be zero. After retaining all old roots except these two, the Schur matrix is $I - t\mathbf{1}\mathbf{1}^T$, $t = (i-1)/(2i)$, positive definite with positive inverse. The replacement is an actual vertex. Its positive displacement has form $Z = W_i\omega_i + W_{i+1}\omega_{i+1}$. The explicit one-helper mixed-point calculation in Sections 4.2–4.3 gives, with $W = W_i + W_{i+1}$,

$$\frac{\sqrt{1+\mathcal{D}}}{W} < \frac{7}{16}, \qquad W_i\sqrt{2i} + W_{i+1}\sqrt{2i+2} - \|Z\| < \frac{W}{4\sqrt{10}}.$$

These use only $i \ge 5$, $A_i > 2$, $\gamma_i > 4/5$ and the displayed Schur matrix. Farthest comparison then makes both indices weak impossible, since $7/16 + 1/(4\sqrt{10}) < .61$. All retained spectator blocks are positive; the same point is feasible for the entire original polytope. Thus weak indices are nonadjacent.

6.1 Strong index five and omitted floors

Suppose first that 5 is strong. At least one of $M, M+1$ is strong. Choose endpoint $K = M$ if possible, otherwise $M+1$. The prefix interpolation through $0, f_5, \dots, f_K$ has initial slope $f_5/5$, subsequent slopes d_i, and zero after K. The slopes are positive and decreasing. Hence its prefix duals are nonnegative, and its squared norm E_K is a lower bound for the actual prefix energy.

Omitting one weak internal floor averages two adjacent slopes. The loss is half a squared second difference, less than $1/3200$: on $t \geq 5$, $|(\sqrt{t})''| < 1/40$. Nonadjacency makes the affected open intervals disjoint. The surviving interpolation stays concave with nonnegative duals, so the losses add even when their closed intervals share an endpoint. At endpoint M there are at most H_M holes. At $M+1$ the count rises by at most one, while

$$E_{M+1} - E_M = d_M^2 > \frac{1}{4M+2} \geq \frac{1}{66} > \frac{1}{3200}.$$

This proves Theorem 6.1 in the strong-five case.

6.2 Weak index five has a better actual lower bound

If 5 is weak, 6 is strong. The complete native pressure table for (30) uses only the first six long roots. Writing $A^2 = 10\gamma^2 + 2\gamma - 1/2$, its actual points give

own label	$u, \quad x \cdot u \geq T$	T
5	$\gamma(\omega_4+\omega_6)/2 + \omega_3/6$	$H(A, 2/3-\gamma^2)$
$7,8$	$\gamma(\omega_4+\omega_6)/2 + \omega_2/4$	$A - \sqrt{\gamma^2 - 3/4}$
≥ 9	$\gamma(\omega_4+\omega_6)/2 + \omega_1/2$	$H(A, 1-\gamma^2)$
dotted	$(\gamma-\gamma^{-1})\omega_5 + (\omega_4+\omega_6)/(2\gamma) + \omega_1/2$	$A - \sqrt{1-\gamma^{-2}}$.

These are the same finite projections and negative-core points constructed in Section 4.2, with the helper reversed and the prefix Gram substituted. Their selected sets are $\{i,1,2\}$, $\{i,1\}$, $\{i\}$ and the dotted pair, respectively. Their coefficients, Gram matrices and time norms do not involve a tail. Positivity gives all unknown wall inequalities, and $Q = 0$ makes the pressure purely in the component.

The coordinate sequence is decreasing. Thus $x \cdot (\omega_4 + \omega_6) \leq 2x \cdot \omega_5$, $x \cdot \omega_2 \leq 2x \cdot \omega_1$, and $x \cdot \omega_3 \leq 3x \cdot \omega_1$. Since 5 is weak, every row implies

$$x_1 > \sqrt{2}\{T - \gamma(\sqrt{10} - .61)\} > .5965.$$

Here are the native scalar seals, retained to make the weak-five boundary explicit. For $5, 7, 8$ the lower bounds exceed $.8, .7, .65$, using

$$.809 < \cos(\pi/5) < .81, \quad .9 < \cos(\pi/7) < .902, \quad .923 < \cos(\pi/8) < .9239.$$

For finite labels at least 9, $H(A, 1-\gamma^2) - \gamma(\sqrt{10} - .61)$ is increasing from $\gamma = .939$: $A' > \sqrt{10}$, $H_A > 9/10$, $H_\delta < 0$. At that endpoint $A > 3.1929$, $\sqrt{A^2+\delta} > 3.211$, $\sqrt{\delta} < .344$, giving $H > 2.88$ and the lower bound above .65. For $1 < \gamma < 6/5$, the dotted difference decreases: $A' < 13/4$, the derivative of $\sqrt{1-\gamma^{-2}}$ exceeds 1, and $\sqrt{10} - .61 > 5/2$. Its endpoint lower bound is

$$\sqrt{2}\{\sqrt{163/10} - \sqrt{11}/6 - (6/5)(\sqrt{10} - .61)\} > .5965.$$

Positive square tests with $\sqrt{163/10} > 4.03732$, $\sqrt{11}/6 < .552771$, $\sqrt{10} < 3.162278$, $\sqrt{2} > 1.41421$ seal it.

Now $S_6 > f_6 > 2.01815$. For a strong endpoint $K = M$ or $M+1$ (use $K = 6$ at $M = 6$), interpolate through

$$(1, .5965), \quad (6, 2.01815), \quad (K, 2.01815 + \sqrt{K} - \sqrt{6}).$$

Omit the repeated last point when $K = 6$. The slopes decrease: $.5965, .28433$, then $1/(\sqrt{K}+\sqrt{6}) < 1/5$. All duals are nonnegative, and its endpoint lies below the actual strong floor. Therefore actual energy exceeds

$$L_M = C + \frac{(\sqrt{M} - \sqrt{6})^2}{M-6}, \qquad C = .5965^2 + \frac{(2.01815 - .5965)^2}{5} = .7600299945,$$

with last term zero at $M = 6$. This expression increases with the endpoint, so $K = M+1$ only helps.

In fact $L_M > E_M$ for the whole range. The identity

$$f_5^2/5 = 1 + \frac{3721}{100000} - \frac{61}{500}\sqrt{10}$$

and $\sqrt{10} > 79/25$ give $f_5^2/5 < 163/250$. Also $\sqrt{30} > 115/21$, since $30 - (115/21)^2 = 5/441$; thus $d_5^2 < 1/21$ and $E_6 < 7/10$. The $k = M - 6 \le 10$ subsequent increments lie in $(1/8, 1/5)$. Their variance is at most

$$\sum d_i^2 - \frac{(\sum d_i)^2}{k} \le \frac{k(1/5 - 1/8)^2}{4} \le \frac{9}{640}.$$

This follows by expanding $(d_i - 1/8)(1/5 - d_i) \ge 0$ and completing the square in their mean. Since their sum is $\sqrt{M} - \sqrt{6}$,

$$L_M - E_M > .76 - .7 - 9/640 > 0.$$

The case $M = 6$ is immediate from $C > .76 > E_6$. This proves Theorem 6.1 without an assumed strong-five bootstrap.

Only two rows are needed for the final bound. The exact rational seal $f_5^2/5 > .6514$ and the increment lower bounds, in units of 10^{-5}, are

$$4545,\ 3846,\ 3333,\ 2941,\ 2631,\ 2380,\ 2173,\ 2000,\ 1851,\ 1724,\ 1612.$$

Each follows from $d_i^2 > 1/(4i+2)$. Adding the appropriate initial segment and subtracting $H_M/3200$ gives

M	B rank	D rank	actual energy
14	≥ 19	≥ 20	$> .90715 > .90$
16	≥ 21	≥ 22	$> .9401975 > .94.$

(31)

Every decimal in these inequalities is an exact rational.

7 Coupling D_{23} to its isolated short spectator

Proposition 7.1. *A whole D_{23} cannot coexist with an isolated short A_1 of type I at a farthest vertex, even with further finite spectators.*

Proof. We first extend the actual prefix window just far enough. For D_{23}, indices $5, \dots, 18$ have right D rank at least 5, so Section 4.1 applies. The weak-five table in Section 6 uses only these long roots. Weak5 would force strong6 and one of16/17 strong, yielding

$$\|x_D\|^2 > .5965^2 + \frac{(2.01815 - .5965)^2}{5} + \frac{1.55051^2}{10} = 1.00043812051 > 1.$$

Endpoint17 only increases the last contribution. Hence5 is strong independently of the tail argument.

At index19 the retained right diagram is D_4. Its complete one-node finite extensions are simple3 at any leaf. A joined left/right bridge has full white type D_{23}, sealing further foreign contacts at its internal new node. Its usual positive endpoint estimate is strong; zero own contact instead gives an affine cycle. The long-leaf row is already bulk. For a fork3 row with no left contact,

$$\delta = \tfrac{1}{2} - Q, \qquad \sigma = \gamma + \tfrac{1}{4}.$$

Own labels $0, 3, 4, 6$ give, respectively, the affine E_8 principal, an all3 cycle, the affine $3, 3, 4, 3$ path, and the affine $6, 3$ path. With $Q > 0$ the ratio $\sigma/\sqrt{\delta}$ is strong. At $Q = 0$, every $\gamma \geq \cos(\pi/7)$ has ratio > 1.61, above the exact edge target < 1.577 at index19. This includes dotted contacts. These are the complete D4-layer cases of Section 4.3; only own5/fork3 with $Q = 0$ needs an additional actual point.

Set $\gamma = \varphi/2$, let f be the contacted fork, and put

$$v = \gamma\omega_{19} + \omega_f/2, \quad A^2 = 19(\varphi + 5/8) - 1/2, \quad \delta_5 = 1 - 3\gamma^2/2.$$

Select the old paths $19, 18, 17$ and $19, 20, 21$. Neither has another contact with c. The two positive H_4 corrections are

$$w_\pm = \gamma(3n_{19}/2 + n_{\text{side}} + n_{\text{second}}/2).$$

Their actual finite projections give, on averaging,

$$y = v - (w_+ + w_-)/2, \qquad x_D \cdot y > A - \sqrt{\delta_5}.$$

The cross product $w_+ \cdot w_- = 3\gamma^2/4$ gives

$$y^2 = A^2 + 1 - 15\gamma^2/8 < A^2, \qquad y \cdot u_5 = \sqrt{10}(\gamma + 1/4).$$

All old corrections have zero first-five coordinates. The exact bounds $A > 6, \|y\| > 6.4, \sqrt{\delta_5} < 1/7$ imply

$$x_D \cdot \widehat{y} > 41/42 > .97, \qquad \widehat{y} \cdot u_5 < 3.3602/6.4 < .55.$$

Strong5 gives $x_D \cdot u_5 > .8$. Their sum then contradicts Cauchy because $1.77^2 > 3.1 > \|u_5 + \widehat{y}\|^2$. Every correction is positive, so these points are feasible against all unknown walls, not just the selected paths.

Thus (30) and its nonadjacency hold through19. One of18/19 is strong. The same prefix reference as before has

$$E_{18} > .97119, \qquad E_{19} > .98470.$$

The three additional increment lower bounds at indices16,17,18 are $1515, 1428, 1351$, in units 10^{-5}; they again follow from $d_i^2 > 1/(4i + 2)$. At most6/7 holes give the common bound

$$\|x_D\|^2 > B, \qquad B = .969315. \tag{32}$$

No index20 or D_3/D_2 tail is used.

If $x_1 \geq a$, with $a = .5$ or $.52$, replace the first five reference slopes by $a, (f_5 - a)/4$ repeated four times. They remain positive and decreasing: $1.8 < f_5 < 1.805$, and the next slope, even after omitting floor6, is $< 1/4$. The exact gain is $(5/4)(a - f_5/5)^2$. Since $f_5/5 < .361$,

$$\|x_D\|^2 > B + \frac{5}{4}(a - .361)^2. \tag{33}$$

Later hole losses and their disjointness are unchanged.

Now let e be the unit old A_1 normal. Its isolated short opposite is

$$c = \sqrt{h}\,p - \sqrt{1+h}\,e, \qquad 0 < h < 1/2.$$

The A_1 edge has $\cosh L = \sqrt{1+h}$, so its projection energy is at least

$$s(h) = \frac{\sqrt{1+h}-1}{\sqrt{1+h}+1}. \tag{34}$$

Consider the actual opposite $d = Ap - v_D - \beta e - v_{\text{other}}$ of direction1 in D_{23}. It differs from c, whose D derivative is zero. If $\beta = 0$, its actual wall product with c and the native gap give $\sqrt{h}\,A \geq 1/2$. If $\beta > 0$, then $\beta \geq 1/2$ and acuteness gives $\sqrt{h}\,A \geq \sqrt{1+h}\,\beta > 1/2$. Thus $A \geq 1/(2\sqrt{h})$ in every case. The unit norm gives $\|v_D\|^2 \leq 1 + A^2$. Since the normalized first D direction is e_1,

$$\tanh L_1 = \frac{A}{v_D \cdot e_1} \geq \frac{A}{\sqrt{1+A^2}}, \qquad x_1 \geq t(h) := \sqrt{1+4h} - 2\sqrt{h}.$$

This bound retains a possible nonzero A_1 contact of d.

The functions s and t increase and decrease, respectively, with exact inverse tests

$$s(h) > u \iff h > \frac{4u}{(1-u)^2}, \qquad t(h) > v \iff h < \frac{(1-v^2)^2}{16v^2}.$$

They divide the whole short-height interval into three cases. For $h \leq 3/25$, $t(h) > .52$: its critical height $12996/105625$ exceeds $3/25$ by $321/105625$. Equation (33) gives $\|x_D\|^2 > 1.00091625$. For $3/25 < h < 2/15$, $t(h) > .5$, since $2/15 < 9/64$, and $s(h) > .02$, since $3/25 > 200/2401$. The orthogonal sum is

$$\|x_D\|^2 + x_{A_1}^2 > .969315 + .02415125 + .02 = 1.01346625.$$

Finally, for $h \geq 2/15$, $s(h) > .031$. Its critical height is $124000/938961 < 2/15$, with positive cross-product difference 17922. Equation (32) then gives total energy > 1.000315. Every interval endpoint is covered. Each case contradicts $\|x\|^2 < 1$, proving the proposition. □

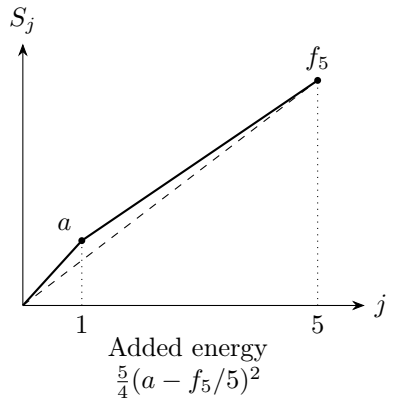
8 The finite forest gives the global bound

We now prove Theorem 1.1 from the component estimates above, without importing a predecessor numerical dimension conclusion. Suppose $25 \leq n \leq 29$. Call an A component of rank at least 7, or B, D of rank at least 9, *dominant*.

If there is none, every rank pays its strict ordinary rank/14 cost, except for the alternatives in (12). Discarding a possible short A_1 gives total energy $> (n-1)/14 > 1$. A rank-two defect gives $> 9/100 + (n-2)/14 > 1$. Thus a dominant block exists. Every dominant block pays more than $1/2$, so there is exactly one. Only now are all its spectators ordinary apart from the single permitted debit. Write its rank as r, their total rank as $S = n - r$. The three spectator lower bounds are

$$\frac{S}{14}, \qquad \frac{S-1}{14}, \qquad \frac{9}{100} + \frac{S-2}{14}. \tag{35}$$

Empty ordinary sums mean zero. In particular, for a sole rank-two defect strictness comes from that block's own $> 9/100$.



$0 < h \le 3/25$: $x_1 > .52$
D energy > 1.00091625

$3/25 < h < 2/15$: $x_1 > .5$
$D + A_1$ energy > 1.01346625

$2/15 \le h < 1/2$: $s(h) > .031$
$D + A_1$ energy > 1.000315

Figure 2: The isolated short wall forces an extra first-coordinate constraint in the same D projection. The left diagram shows the reference adjustment; the right lists the exact height cases. The height boxes are not drawn to scale.

For a dominant A, the half bound first gives $S \le 7$. Indeed $S \ge 7$ contradicts the norm without a debit, and $S \ge 8$ does so with either debit. Hence $r \ge 18$, so the stronger $5/7$ estimate applies. A second use of (35) gives $S \le 4$: the ordinary threshold is $S = 4$, the short-A_1 threshold is $S = 5$, and the rank-two threshold at $S = 5$ pays

$$5/7 + 9/100 + 3/14 = 713/700 > 1.$$

Thus $r = n - S \ge 21$, contrary to the A cap20.

For dominant B, D, the $9/14$ floor gives $S \le 5$ and therefore $r \ge 20$. The caps leave $r \le 22$ for B and $r \le 23$ for D. If $r = 20, 21$, (31) gives energy $> .90$ and $S \ge 4$. Even after discarding a short A_1, three ordinary ranks pay $> 3/14 > .10$; a rank-two defect is stronger. If $r = 22, 23$, the energy is $> .94$ and $S \ge 2$. One retained ordinary rank pays $> 1/14 > .06$; if instead the spectators are solely a defect, that block pays $> .09 > .06$. The fully ordinary cases are stronger. All possibilities contradict (1).

Dimensions 25 through 29 are therefore impossible. Vinberg's published ceiling29 excludes all higher dimensions [1]. It remains to exclude dimension24.

Lemma 8.1 (The dimension-twenty-four frontier)**.** *At a farthest vertex in dimension24, the preceding bookkeeping leaves only A_{20} with four spectator ranks and a permitted debit, or D_{23} with one isolated short A_1 of type I.*

Proof. Existence and uniqueness of a dominant component follow by the same norm calculation, which already holds in dimension24. For a dominant A, the half bound gives $S \le 7$ and hence $r \ge 17$. The $5/7$ bound then gives $S \le 4$, so the cap20 forces $r = 20, S = 4$. Fully ordinary spectators would already give energy $> 5/7 + 4/14 = 1$, leaving a permitted debit. For dominant B, D, $S \le 5$ gives $r \ge 19$. For B, D ranks $19, 20, 21, 22$, the generic theorem's respective prefix bounds are $> .88864, .90715, .9240775, .9401975$; these and the remaining ordinary ranks exclude all their debit cases. The B cap22 and D cap23 leave only $D_{23}, S = 1$. An ordinary A_1 costs more than the remaining .06. A rank-two defect and type J both require two ranks, so the sole spectator is isolated type I. □

The first frontier is excluded by Proposition 5.2. After discarding one short A_1, its other three spectator ranks still pay $> 3/14$, so total energy is $> 11/14 + 3/14 = 1$. For a rank-two defect, the

cost is

$$11/14 + 9/100 + 2/14 = 713/700 > 1.$$

The other frontier is exactly Proposition 7.1. Thus dimension24 is impossible too. This completes Theorem 1.1. It makes no claim of sharpness, an improved arithmetic bound, or a new compact example.

A Supporting derivations and verification scope

The proof above integrates the supporting written arguments into one exposition. The following correspondence is a guide to those derivation families, not a list of additional mathematical assumptions.

Part of this paper	Supporting derivation family
Sections 2–3	Actual mixed replacements and reciprocal shadows; universal small-component costs; the short-A_1 and isolated rank-two debit arguments.
Proposition 4.1	Complete low-index common-pair lists, the native-seven actual dual foot, and the fixed positive three-pair inequality.
Proposition 4.2	Foreign-pressure clearing, bulk and short-tail continuation, actual two-cap replacements, and the A_{21} plateau certificate.
Section 5	Rank-seven base, uniform half and five-sevenths pressures, and the stronger A_{20} four-foot argument.
Sections 6–7	Generic classical prefix interpolation and the coupled isolated-A_1 obstruction.
Section 8	The dimension-24 frontier and its two final exclusions.

Exact source versions, individual contribution statements, mathematical review records and source-integrity records are preserved in the project proof map, `docs/compact_farthest_vertex_bounds_proof_map_20261020.md`, and its immutable source packet, `results/compact_farthest_vertex_dimension23_packet_20261020.json`. No earlier packet or review is changed by the present integration. Several feet and interpolation estimates were derived jointly before outside-author reading; repeated derivations are not counted as independent discoveries.

The accompanying proof-assistant modules verify selected accounting implications with the component estimates as explicit premises. They do not formalize the Lorentz geometry, the finite continuation classifications, or the component energy proofs. The proof of the theorem is the written argument above, from the geometric and classification inputs stated in Section 1; source integrity checks and typesetting serve different purposes.

References

[1] E. B. Vinberg, *The absence of crystallographic groups of reflections in Lobachevsky spaces of large dimension*, Trudy Moskov. Mat. Obshch. **47** (1984), 68–102; English translation, Trans. Moscow Math. Soc. (1985), 75–112. Primary bibliographic record.

[2] E. B. Vinberg, *Hyperbolic reflection groups*, Russian Math. Surveys **40**:1 (1985), 31–75, doi:10.1070/RM1985v040n01ABEH003527.